

Review

Carbon Dioxide Activation and Hydrogenation into Value-Added C1 Chemicals over Metal Hydride Catalysts

Malesela A. Mafokoane ^{1,2,3} , Xiaoxia Ou ^{2,3,*}, Nicholas M. Musyoka ^{2,3,*} and Fei Chang ^{1,*} 

¹ Yongjiang Laboratory, Ningbo 315201, China

² Nottingham Ningbo China Beacons of Excellence Research and Innovation Institute, University of Nottingham Ningbo China, Ningbo 315100, China

³ Faculty of Science and Engineering, University of Nottingham Ningbo China, Ningbo 315100, China

* Correspondence: xiaoxia.ou@nottingham.edu.cn (X.O.); nicholas.musyoka@nottingham.edu.cn (N.M.M.); fei-chang@ylab.ac.cn (F.C.)

Abstract: The utilisation of fossil fuels has resulted in the continuous increase in anthropogenic carbon dioxide (CO₂) emissions and has led to significant environmental impacts. To this end, the catalytic hydrogenation of captured CO₂ into value-added C1 chemicals has attracted great attention. In this case, significant research efforts have been directed towards the development of heterogeneous catalysts. Owing to the unique properties and functionalities of hydridic hydrogen (H[−]), metal hydrides have shown great promise in hydrogen-involved catalytic processes. This is attributed to their enhanced hydrogen (H₂) absorption-desorption reversibility and newly developed active sites. Nevertheless, their application in the activation and hydrogenation of CO₂ has been overlooked. In this review paper, we provide an overview of recent advances in catalytic CO₂ hydrogenation using metal hydride-based materials. Firstly, the reaction mechanisms of CO₂ hydrogenation toward different C1 products (CO, CH₄, CH₃OH and HCOOH) are introduced to better understand their application trend. Thereafter, we highlight the challenges of developing robust hydride catalysts with different components and structures that enable tuning of their catalytic activity and selectivity. A brief introduction of the CO₂ hydrogenation over typical homogeneous metal hydrides complexes is also presented. Lastly, conclusion, future outlook and perspectives are discussed.

Keywords: CO₂ hydrogenation; metal hydrides; H₂ dissociation; C1 chemicals



Academic Editors: Maria A. Goula and Maria Cornelia Iliuta

Received: 2 April 2025

Revised: 20 April 2025

Accepted: 22 April 2025

Published: 26 April 2025

Citation: Mafokoane, M.A.; Ou, X.; Musyoka, N.M.; Chang, F. Carbon Dioxide Activation and Hydrogenation into Value-Added C1 Chemicals over Metal Hydride Catalysts. *Catalysts* **2025**, *15*, 424. <https://doi.org/10.3390/catal15050424>

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

Human-induced activities have resulted in increased greenhouse gas (GHG) emissions, particularly CO₂, which have led to the accelerated rate of global warming beyond the pre-industrial stage [1]. The continuous releases of CO₂ emissions have a significant effect on the earth's natural carbon cycle. The increased concentration of CO₂ in the atmosphere results in great environmental impacts such as global warming, rises in sea level, ocean acidification and climate change [2]. The transformation of CO₂ by using renewable green hydrogen into sustainable and value-added C1 chemicals such as methanol (CH₃OH), formic acid (HCOOH) and methane (CH₄) can be an ideal solution to mitigate the emissions of CO₂, depleting fossil resources, and maintaining energy security. Due to its strong thermodynamic stability and kinetic inertness, the non-polar covalent CO₂ has a very low conversion efficiency and relatively high water solubility (1.45 g L^{−1}) at standard temperature and pressure [3]. This has led to extreme difficulty in CO₂ activation and conversion to value-added hydrocarbons [4]. The catalytic hydrogenation of greenhouse

CO₂ to C1 chemicals is a promising route to store energy within chemical bonds and/or convert it into green, renewable chemicals [5].

Catalysts play an important role in promoting the activation and conversion of CO₂ and dissociation of hydrogen (H₂), and thus yielding desired products, particularly C1 chemicals. This is based on the recent studies reported using different catalysts [6–8]. The C1 chemicals find applications in various fields, such as transport, textile, agriculture, pharmaceutical, petroleum, and the food industry, and in particular, as feedstock for the manufacturing of pharmaceuticals, plastics, leathers, dyes, rubbers etc. (Figure 1) [9–12]. They also play an important role as a potential clean energy carrier and storage, and substitute fuels in fuel cells, ships, and vehicles [13]. In heterogeneous catalysis, the reaction mechanism can be influenced by the different exposure of catalyst surfaces, defects, or electronic structures [14,15]. In recent years, in situ experimental and advanced theoretical studies have given advances in the identification and presentation of mechanisms in the CO₂ hydrogenation reactions into C1 chemicals. The simplified reaction pathways for CO₂ hydrogenation to different C1 products, i.e., methane, formic acid, carbon monoxide and methanol have been summarised in Figure 2 [16]. On the other hand, the reaction equations and thermodynamic data of CO₂ hydrogenation to produce different C1 chemicals are described by Equations 1 to 5 [15,17–22].

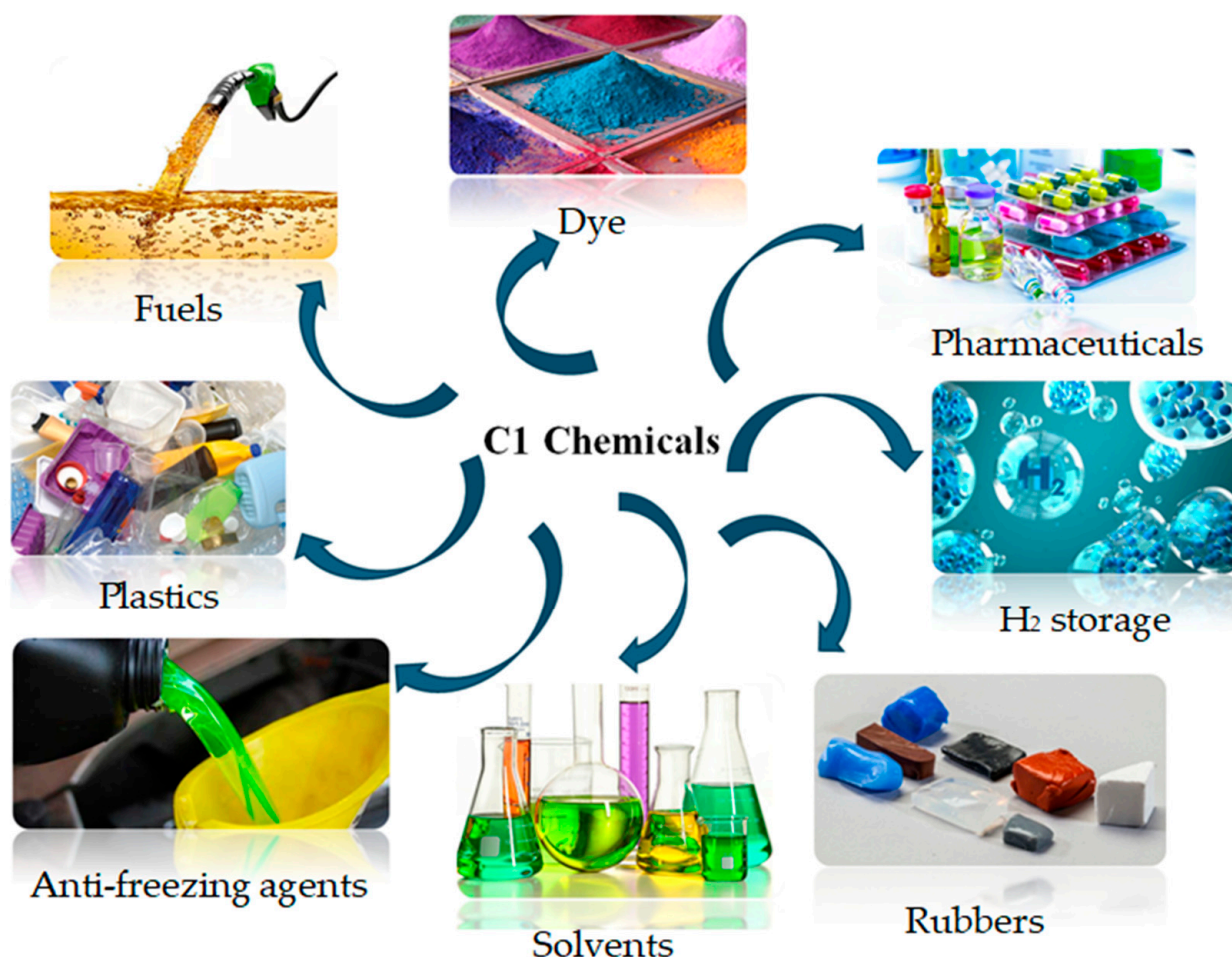


Figure 1. Schematic diagram for the applications of C1 chemicals.

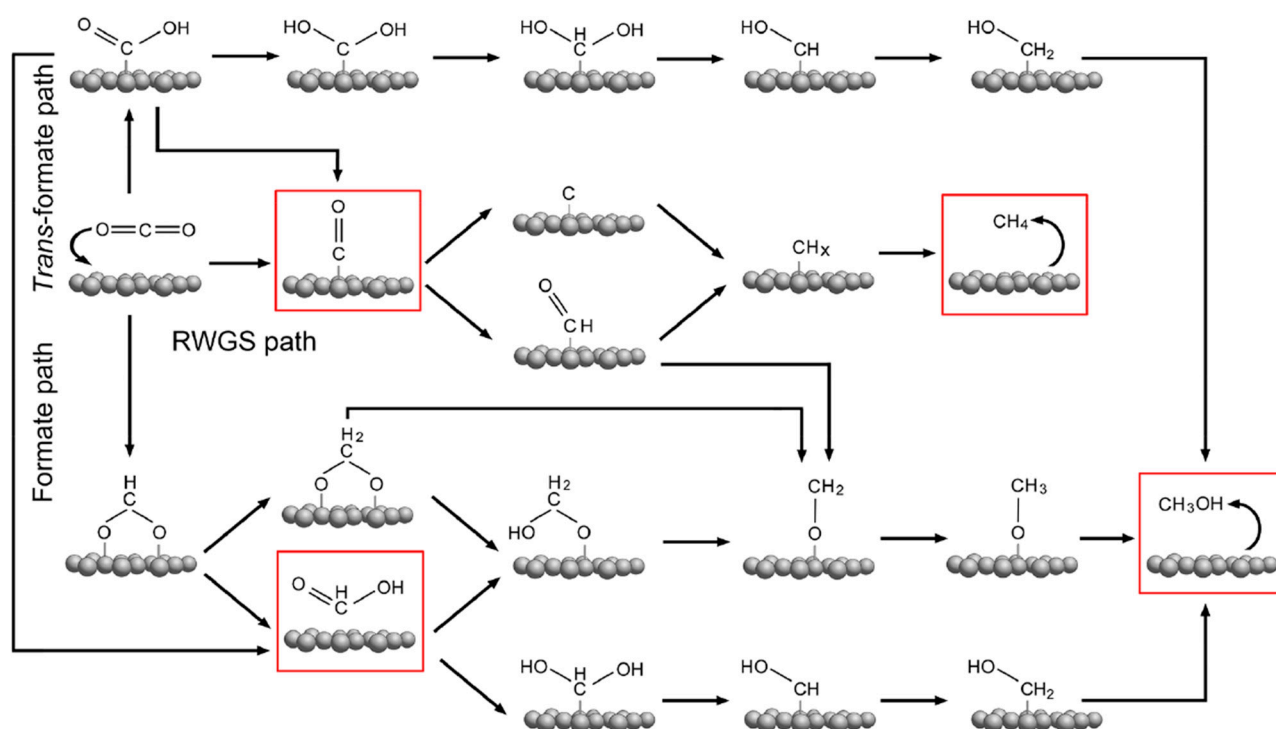
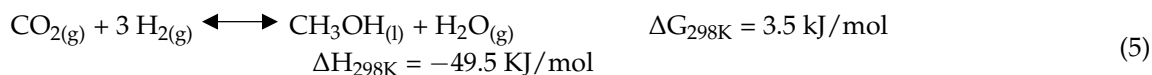
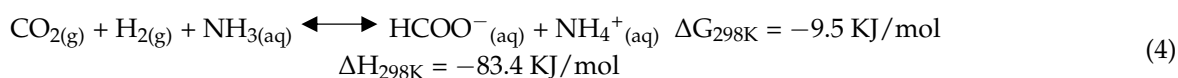
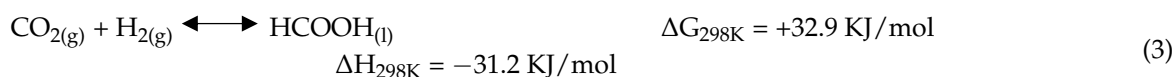
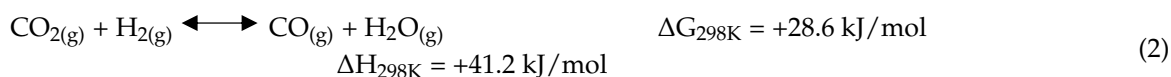
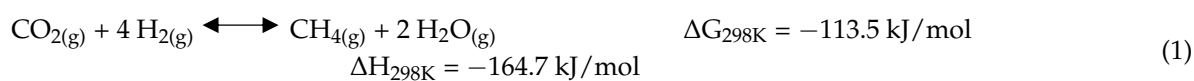


Figure 2. Reaction mechanisms of CO₂ hydrogenation to C1 chemicals: methane, carbon monoxide, formic acid and methanol. Note: the C1 chemicals are in a highlighted red box. Adapted from Wang et al. [16].



During the catalytic process on a solid catalyst surface, the CO₂ molecules undergo adsorption, dissociation, and bond cleavage. Since the direct dissociation of CO₂ into intermediate species such as CO and O atoms [23,24] is one of the key steps in the hydrogenation reaction into C1 chemicals such as methane and methanol [16], it is important to select a functional catalyst with a surface that can adsorb CO and O adequately. The catalyst must also promote the hydrogenation of the dissociated intermediate species into the desired products. The surface should not have a strong binding force to avoid surface poisoning and further prevent the adsorption of CO₂ [25,26]. For example, the Cu(110) surface exhibited the most activity towards CO₂ dissociation into CO and atomic O. However, the adsorbed atomic oxygen was found to poison and block the surface, leading to the inhibition of further CO₂ adsorption [25]. In contrast, by increasing the catalytic reaction temperature and applying a molar ratio of CO₂:H₂ of 1:1, the atomic O adsorbed

at the surface of Pt(111) favoured the reverse water gas shift (RWGS) reaction due to the reduction in the dissociated O₂ [26]. Metal hydrides are starting to emerge and showing significant improvements in the hydrogen-involved catalytic process such as ammonia synthesis, alkyne hydrogenation and carbon dioxide methanation. In the next chapter we will highlight the properties of metal hydrides and their applications in hydrogenation reactions.

2. Metal Hydrides

2.1. Basic Properties of Metal Hydrides

Metal hydrides are compounds that consist of metal cations and hydride anions. When pressurised, metals bind with hydrogen to form stable metal hydrides, which can be used for convenient hydrogen storage [27]. Metal hydrides have been known throughout history dating back to the 19th century. Potassium hydride was the first to be synthesised in 1811 by chemically reacting and heating metal potassium with gaseous hydrogen molecules [28]. The equilibrium pressure plateau for the alkali metals (potassium, sodium and lithium) reacting with hydrogen was first measured in 1874 [29]. Metal hydrides possess reversible hydrogen absorption-desorption ability under suitable reaction conditions [30,31]. As a result, they can be applied in H₂ storage and in other fields such as in hydrogen-involved catalytic reactions [32]. Metal hydrides play a fundamental role in the environmental and energy catalytic reactions associated with hydrogenolysis, dehydrogenation, reforming, and catalysed hydrogenation [30]. They chemically bind hydrogen and form a dynamic boundary interface. This allows hydrogen to be absorbed, diffuse and interacts with the metal lattice. The abundant bonding sites on metal hydride's surface and bulk phase distinguishes them from conventional metal catalysts. Furthermore, amongst the advantages of nanostructured metal hydrides in comparison to bulk catalysts are their tuneable heats of absorption-desorption of hydrogen, enhanced hydrogen absorption-desorption reversibility, and newly developed active surface sites proficient in activating stable chemical bonds and accelerate reaction rates [33]. Attributed to the differences and variations in electronegativities between metal elements and hydrogen, metal hydrides have been reported to have different types of hydrogen bonding and are classified into five categories (Table 1). In a recent review article, a summary of the different types of metal-hydrogen bonding was discussed, highlighting the different classification of metal hydrides [31]. It is noteworthy that the different classes are not fixed; some of the metal hydrides can merge between two classes.

Table 1. Classification of metal hydrides by metal-hydrogen bonding with some selected examples.

Category	Metal Hydride
Ionic hydrides	LiH, NaH, BaH ₂ , CaH ₂ , MgH ₂
Interstitial hydrides	YH ₃ , PdH _x , FeTiH _{1.7} , LaNi ₅ H ₆ ,
Covalent hydrides	AlH ₃ , CuH, NH ₃ BH ₃ , BeH ₂
Complex hydrides	Mg ₂ NiH ₄ , LiAlH ₄ , NaBH ₄ , BaReH ₉
High-pressure metal hydrides	Li ₆ NbH ₁₁ , Li ₅ MoH ₁₁ , Mg ₃ CrH ₈

Ionic, interstitial and covalent hydrides have been grouped and regarded as binary and/or intermetallic hydrides by other researchers. On the other hand, Wietelmann et al. [34] classified the metal hydrides based on the metal's position on the periodic table (Figure 3). Each category of metal hydrides has been reported to exhibit varying strengths and weaknesses. Refer to Table 2 for a summary of the benefits and challenges of the various categories of metal hydrides [33,35].

	Ionic saltlike
	Metallic interstitial
	Covalent polymeric
	Covalent

1	2											13	14	15	16	17
LiH 0.97	BeH ₂ 1.47											BH ₃ 2.01	CH ₄ 2.50	NH ₃ 3.07	H ₂ O 3.50	HF 4.10
NaH 1.01	MgH ₂ 1.23	3	4	5	6	7	8	9	10	11	12	AlH ₃ 1.47	SiH ₄ 1.74	PH ₃ 2.06	H ₂ S 2.44	HCl 2.83
KH 0.91	CaH ₂ 1.04	ScH ₂ 1.20	TiH ₂ 1.32	VH VH ₂ 1.45	CrH (CrH ₂) 1.56	MnH _{c1} 1.60	(FeH _{c1}) 1.64	(CoH _{c1}) 1.70	NiH _{c1} 1.75	CuH 1.75	ZnH ₂ 1.66	(GaH ₃) 1.82	GeH ₄ 2.02	AsH ₃ 2.20	H ₂ Se 2.48	HBr 2.74
RbH 0.89	SrH ₂ 0.99	YH ₂ YH ₃ 1.11	ZrH ₂ 1.22	(NbH) (NbH ₂) 1.23	(MoH _{c1}) 1.30	TcH _{c1} 1.36		(RhH _{c1}) 1.45	PdH _{c1} 1.35	(AgH) 1.42	(CdH ₂) 1.46	(InH ₃) 1.49	SnH ₄ 1.72	SbH ₃ 1.82	H ₂ Te 2.01	HI 2.21
CsH 0.86	BaH ₂ 0.97	LaH ₂ LaH ₃ 1.08	HfH ₂ 1.23	TaH _{c0.5} 1.33	(ReH _{c0.2}) 1.46				(PtH _{c1}) 1.44	(AuH) 1.42	(HgH ₂) 1.44	(TlH ₃) 1.44	PbH ₄ 1.55	BiH ₃ 1.67	H ₂ Po 1.76	HA 1.96
		AcH ₂ 1.00														
Lanthanides		CeH ₂ 1.06	PrH ₂ PrH ₃ 1.07	NdH ₂ NdH ₃ 1.07		SmH ₂ SmH ₃ 1.07	EuH ₂ 1.01	GdH ₂ GdH ₃ 1.11	TbH ₂ TbH ₃ 1.10	DyH ₂ DyH ₃ 1.10	HoH ₂ HoH ₃ 1.10	ErH ₂ ErH ₃ 1.11	TmH ₂ TmH ₃ 1.11	(YbH ₂) YbH ₃ 1.06	LuH ₂ LuH ₃ 1.14	
Actinides		ThH ₂ 1.11	PaH ₂ 1.14	UH ₃ 1.22	NpH ₂ NpH ₃ 1.22	PuH ₂ PuH ₃ 1.22	AmH ₂ AmH ₃ 1.2									

Figure 3. Periodic table classifying metal hydrides according to bonding types. Note: The hydrides that are unstable at ambient temperatures are in parentheses and are formed at very high H_2 pressure. Adapted from Wietelmann et al. [34].

Table 2. A generalised summary of the drawbacks and benefits of categories of the metal hydrides. Note MHx: M = main group or transition metal; ABxHy: A = hydriding metal, B = non hydriding metal; MEHx: E = boron, nitrogen or aluminium [33,35].

Category	Metal hydride	Drawbacks	Benefits
Binary	MH _x , NaH, LiH, MgH ₂ , CaH ₂	- Strong ionic metal-hydride bond - High energy cost to desorb the hydridic hydrogen	Highly down-sizeable to change the thermodynamics of the H₂ absorption.
Intermetallic	AB _x H _y TiMn ₂ H _x , TiFeH _x , LaNi ₅ H _x	- Limited H₂ by weight storage of most AB_xH_y (of 1–2% except Mg alloys) - Relatively lower absorption and storage of H₂ per mass	- High molar H₂ absorption - Faster H₂ absorption-desorption kinetics when H₂ is stored in the interstitial sites of metal lattice.
Complex	MEH _x NaAlH ₄ , LiBH ₄ , LiNH ₂ , Mg(BH ₄) ₂ , Mg(NH ₂) ₂ , Mg(AlH ₄) ₂	- Slower H₂ absorption-desorption kinetics - Limited and poor reversibility - Complex reaction pathways - Strong chemical bonding - Highly endothermic during hydrogen desorption or release - High activation energy and desorption temperature	- Higher hydrogen content by weight - Diverse adjusting space of thermodynamic properties toward hydrogenation and dehydrogenation

Higher temperatures are required to break and dissociate the metal-hydrogen bond during desorption of hydrogen or hydrogenation. This is attributed to the covalent and ionic bonds between the transition and main group metals with hydrogen [35]. Moreover, transition metals such as platinum group (Ir, Pt, Pd and Rh) metals were noted for their reversible hydrogen absorption-desorption at ambient temperature due to their higher density states at the Fermi level [36]. During the hydriding process, the dissociative adsorption of H_2 on the metal oxide (M-O) heterolytically splits H_2 into metal hydride (M-H) and O-H (hydroxide) [37,38]. Similarly, Serykh [39] demonstrated that during the dissociative adsorption of H_2 on Al_2O_3 -supported In_2O_3 , H_2 splits into and forms indium hydride (In-H) and hydroxyl (O-H) sites on the In_2O_3/Al_2O_3 . A better discussion and illustration of the hydride formation on supported and unsupported metal oxides will be reported later in this review. Although a variety of metal hydrides are regarded as basic and nucleophilic, the hydrides of group 13 metals from Figure 3 are electron deficient and stronger Brønsted acids [34]. Analogous to metal hydrides, metal oxyhydride is a form of a hydride where a metal oxide is bonded to hydrogen through the metal center with a similar nature of chemical bonding to the complex hydride [31]. Amongst the different anion dopants (H^- , N_3^- , F^- , e^- , and S_2^-) to modify the electronic, physical and chemical structure of the oxide, H^- is a better dopant and/or substitute for the O_2^- sites. This is attributed to a similar ionic radius between O_2^- (135 pm) and H^- (134 ± 2 pm), and one additional electron which maintains the lattice charge neutrality [40]. Therefore, the most important step during the catalytic hydrogenation reaction on metal hydrides is the transfer of hydride anion from the parent metal hydride complex to the unsaturated substrate, such as CO_2 , alkyne, and N_2 [41].

Hydrogen-involved catalysis over metal hydrides has been reported in recent years, principally on the alkyne hydrogenation, and ammonia synthesis reactions. For example, the Pd phase transition between α - and β - PdH_x/ZnO was evidenced from Pd/ZnO during the acetylene hydrogenation to exhibit excellent catalytic activity [42]. The role of hydride at the subsurface of Pd was studied by Teschner et al. [43] for the reduction of 1-Pentyne. As mentioned above, both selective and non-selective hydrogenation reactions were observed due to bulk properties decoupling from the Pd subsurface and the presence of high hydrogen β -hydride, respectively. A 5 wt.% PdH_x/Al_2O_3 was applied on a liquid phase furfural hydrogenation [44]. Although the hydride was quickly consumed and reduced to metallic Pd and Pd carbide, it was revealed that the hydride phase (bulk and surface hydrogen atoms) was easily consumed only in the presence of furfural as a reactant. However, when cyclohexene was used, the bulk hydrogen atoms of $PdH_{0.4}$ phase were not consumed. The activation and hydrogenation of N_2 have also received attention in recent years using alkali, alkaline earth and transition metal hydrides as heterogeneous catalysts in ammonia synthesis [45–49].

The dissociated lattice H^- from the MgH_2 metal hydride enhanced the hydrogenation of CO_2 by interacting with the C of the CO_2 to form Mg formate as an intermediate, and subsequently hydrogenating into C_2^- - C_4^- lower olefins [50]. Metal hydrides based on Mg were applied in the hydrogenation of dibenzyltoluene and exhibited the best catalytic activity which was attributed to the presence of formed superficial active centers of $MgNiH_4$ on the matrix of MgH_2 [51]. The hydrogenation process is a selective reaction on the surface of a heterogeneous catalyst.

The extent and type of the substrate's reduction depends heavily on the functional groups of the reactant involved such as alkene $C=C$, alkyne $C\equiv C$ triple bond, carbonyl $C=O$, and $N\equiv N$ triple bond. Based on the above-mentioned reactions using metal hydrides, inspiration can be drawn to synthesise a novel metal hydride that can actively hydrogenate CO_2 . As a result, it can be concluded that metal hydrides could be applied in CO_2 reduction

to C1 chemicals, namely, methane, methanol and formic acid. This can result in abundant CO₂ resource utilisation and environmental sustainability by continuously mitigating the excessive emissions. Moreover, this can replace the depleting fossil carbon resources and sustain the energy security. The advances in CO₂ hydrogenation using metal hydride catalysts will be discussed in the next section.

2.2. CO₂ Hydrogenation over Metal Hydrides

2.2.1. Synthesis of CH₄ over Metal Hydrides

The formation of metal hydrides involves the conventional synthesis of a metal precursor followed by a hydriding process. Table 3 lists the metal hydrides employed in the hydrogenation of CO₂ to a C1 chemical (methane).

Table 3. Metal hydrides employed in the catalytic hydrogenation of CO₂ to methane.

Type of Catalyst	Hydriding Conditions	Reaction Conditions	XCO ₂ [%]	CH ₄ Y [%]	CH ₄ [g/kg.h]	CH ₄ S [%]	Ref
Mg ₂ NiH ₄	T = 250 °C, P = 2 MPa H ₂ , t = 48 h Magnetron Sputtering	TOS = 6.5 h, P = 10 bar, T = 330–600 °C, (H ₂ :CO ₂ = 6:1), WHSV = 1200 mL/gcat.h	100	80			[52]
Mg ₂ NiH ₄	T = 250 °C, P = 2 MPa H ₂ , t = 48 h Magnetron Sputtering	TOS = 6.5 h, P = 1 bar, T = 330–600 °C, (H ₂ :CO ₂ = 6:1), WHSV = 1200 mL/gcat.h	100	75			[52]
MgH ₂ - 5wt.%Ni- 5wt.% CNT	Milling ^b , t = 5 h, 500 rpm, 53:1 Ball to powder	TOS = 48 h, T = 350 °C, (MgH ₂ :CO ₂ = 4:1)	100	79		70.9	[53]
Ni- MgO/MgH ₂	Milling ^b , t = 3–15 h, 270 rpm, 47:1 Ball to powder	TOS = 5 h, P = 1 MPa, T = 380 °C, (H ₂ :CO ₂ = 4:1), WHSV = 9600 mL/gcat.h	85.2		944.6	99.5	[54]
Mg ₂ NiH ₄	Milling ^b , t = 60 h, 500 rpm, 10:1 Ball to powder, P = 20 bar H ₂	TOS = 5 h, P = 0.12 MPa, T = 400 °C, (H ₂ :CO ₂ = 4:1)	100	95			[55]
LaNi ₅ H ₅	Heating, t = 5 h, T = 150 °C, P = 4 MPa H ₂	TOS = 400 h, P = 3 MPa, T = 200 °C, (H ₂ :CO ₂ = 4:1)	99	72			[56]
Al-MgH ₂	Milling ^c , t = 5 h, 270 rpm, T = amb, P = 1 MPa H ₂	TOS = 2 h, P = 1 MPa, T = 320 °C, (H ₂ :CO ₂ = 5:1), WHSV = 4800 mL/gcat.h	27.1			88.4	[57]
MgH ₂ /Cu _x O	Milling ^c , t = 3 h, 270 rpm, T = amb, P = 1 MPa H ₂	TOS = 2 h, P = 1 MPa, T = 300 °C, (H ₂ :CO ₂ = 10:1), WHSV = 2400 mL/gcat.h	20.7			90	[50]
MgH ₂ - 10wt% Co	Milling ^b , t = 5 h, 500 rpm, 53:1 Ball to powder	TOS = 48 h, T = 350 °C, (MgH ₂ :CO ₂ = 4:1)		78			[58]
Mg ₂ FeH ₆	Milling ^b , t = 5 h, 500 rpm, P = 1 MPa H ₂ 10:1 Ball to powder	TOS = 5 h, T = 400 °C, (H ₂ :CO ₂ = 4:1)	100	a			[59]
Mg ₂ NiH ₄	Milling ^b , t = 5 h, 500 rpm, P = 1 MPa H ₂ , 10:1 Ball to powder	TOS = 10 h, T = 400 °C, (H ₂ :CO ₂ = 4:1)	100	a			[59]
XH ₂ (X= Mg or Ca)	Milling ^b , t = 48 h, 90:1 Ball to powder	TOS = 48 h, P = 0.25 MPa, T = 450 °C, (XH ₂ :CO ₂ = 2:1)		>80			[60]

a = strong FTIR bands of CH₄ and weak CO, b = mechanical ball milling, and c = reactive ball milling, TOS = Time on stream, WHSV = weight hourly space velocity, X = conversion, Y = yield, and S = selectivity.

Kato et al. [61] studied the behavior of H₂ desorption on bimetallic hydride ZrCoH_x and elucidated its catalytic activity in CO₂ hydrogenation. The atomic flux of hydrogen

desorbed from ZrCoH_x facilitated CO_2 reduction at the interfacial sites (Figure 4a) with a linear proportionality between the total amounts formed CH_4 and hydride concentration in ZrCoH_x (Figure 4b). The diffusion of hydrogen atoms from the metal hydride enhanced the CO_2 methanation. This behavior was not found on the pristine intermetallic ZrCo .

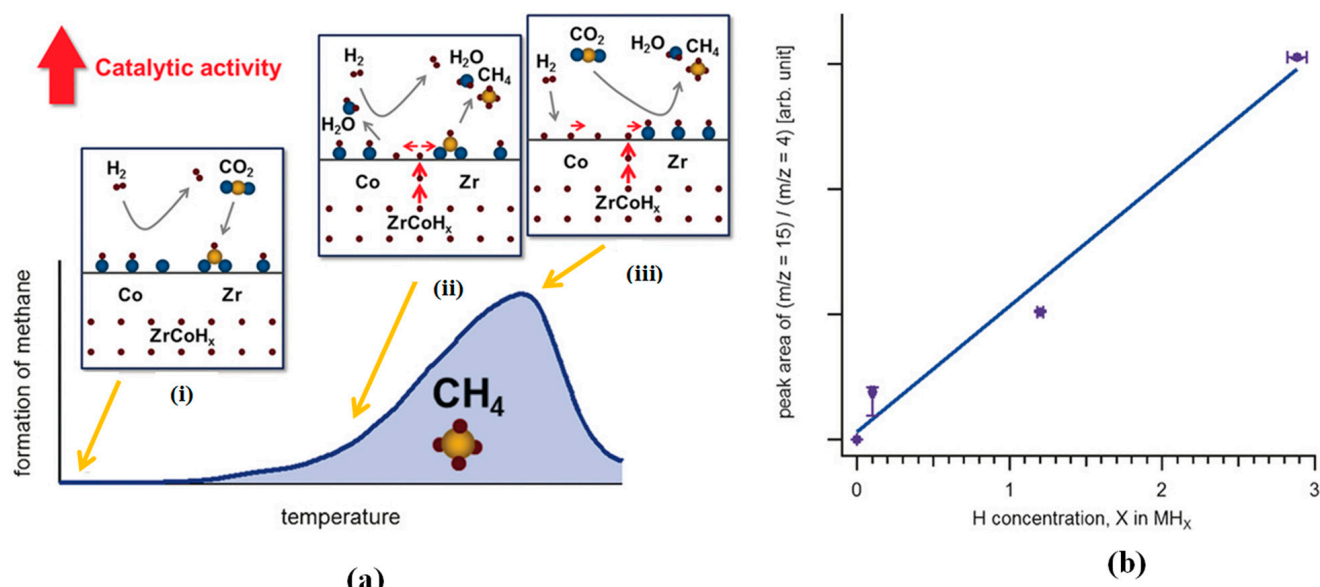


Figure 4. (a) The schematic depiction of the CO_2 reduction over ZrCoH_x . (i) The surface coverage of zirconia by CO_2 to form formates, (ii) the diffusion of H atoms from the ZrCoH_x reacting with formate intermediate producing initially H_2O and CH_4 , and simultaneous reduction of cobalt (Co), (iii) H_2 molecules impinging on the interface dissociates on the reduced Co to form H atoms, and together with the H atoms diffusing from the metal hydride spill over the formate to generate H_2O and CH_4 . (b) The peak area of ion current ratio ($m/z = 15$ (CH_3^+)) : $m/z = 4$ (carrier gas He^+)) as a function of the initial concentration of hydrogen atom in ZrCoH_x ($x = 0, 0.1, 1.2, 2.9$). Adapted from Kato et al. [61].

In another example, Ni hydride as an active phase, helped to dissociate the highly absorbed CO_2 molecules on the surface of Mg_2NiH_4 hydride, leading to a selective formation of CH_4 [62]. In this case, the dissociative adsorption of CO_2 on the hydride was the rate determining step of methanation. In a different study that utilised Mg_2NiH_4 hydride catalysts for the Sabatier type methanation reaction [52], it was found that whilst there were differences in the crystallisation of MgNi_2 and MgNiH_4 (low-temperature monoclinic and high-temperature pseudo-cubic phases), heating at 330–380 °C under H_2 and CO_2 gas flow initiated Mg_2NiH_4 hydride decompositions, disproportionation, and oxidation, empowering and promoting the methanation of CO_2 . In addition, CO_2 methanation at 10 bars started at 50 °C lower than under 1 bar reaching 100% CO_2 conversion and 80% CH_4 selectivity at 420–430 °C. This demonstrates that a high H_2 partial pressure is a stronger factor for the Mg_2NiH_4 stabilisation over the oxidative potential by CO_2 . A study comprising both an experimental and DFT calculations was conducted to evaluate the changes in the structure and reduction mechanism of Ni-incorporated MgO/MgH_2 for CO_2 methanation [54]. A very high CH_4 selectivity of 99.5% and space-time yield of 944.6 $\text{g kg}^{-1}\text{h}^{-1}$ were obtained at 300 °C and 1 MPa $\text{N}_2/\text{CO}_2/\text{H}_2$. These results corroborated the theoretical DFT studies. They both suggested that the formation of H^+ (from the Ni incorporated in $\text{MgO}(110)$) is necessary to activate the terminal O from the CO_2 to form CO^* species. Meanwhile, the hydridic hydrogen (H^-) from MgH_2 hydrogenates the terminal C of the CO_2 to CH_4 . In a different study, carbon nanotubes (CNTs) served as H^- protector of MgH_2 -5wt.%Ni-5wt.%CNT preventing it from oxidation during the CO_2 reduction at different reaction

temperatures [53]. This enhanced the catalytic activity of the metal hydride catalyst leading to a CH₄ yield and selectivity of 79 and 70.9%, respectively, at 350 °C. High catalytic CO₂ conversion (100%) and CH₄ selectivity (95%) of cubic Mg₂NiH₄ complex hydride were achieved at 400 °C in a time of stream of 5 h, and its activity was strongly dependent on the ball-milling time before the CO₂ reduction [55].

Evenly dispersed Ni nanoparticles and a shortened distance of hydridic hydrogen H[−] mobility on the LaNi₅H₅ hydride surface led to a superior catalytic activity of 99% CO₂ conversion and 72% CH₄ yield at a time of stream of 400 h (Figure 5) [56]. In addition, the presence of La in LaNi₅H₅ hydride and the lowered reaction temperature inhibited the agglomeration of Ni nanoparticles (NPs). The basic nucleophilic H[−] of the metal hydride can activate the acidic and electrophilic CO₂ during catalytic reactions. The dissociated lattice H[−] from the Al-doped MgH₂ hydride nano-catalyst achieved 27.1% of CO₂ conversion and 88.4% of CH₄ selectivity under the reactions conditions at 320 °C and 1 MPa of 5:1 H₂/CO₂ ratio [57].

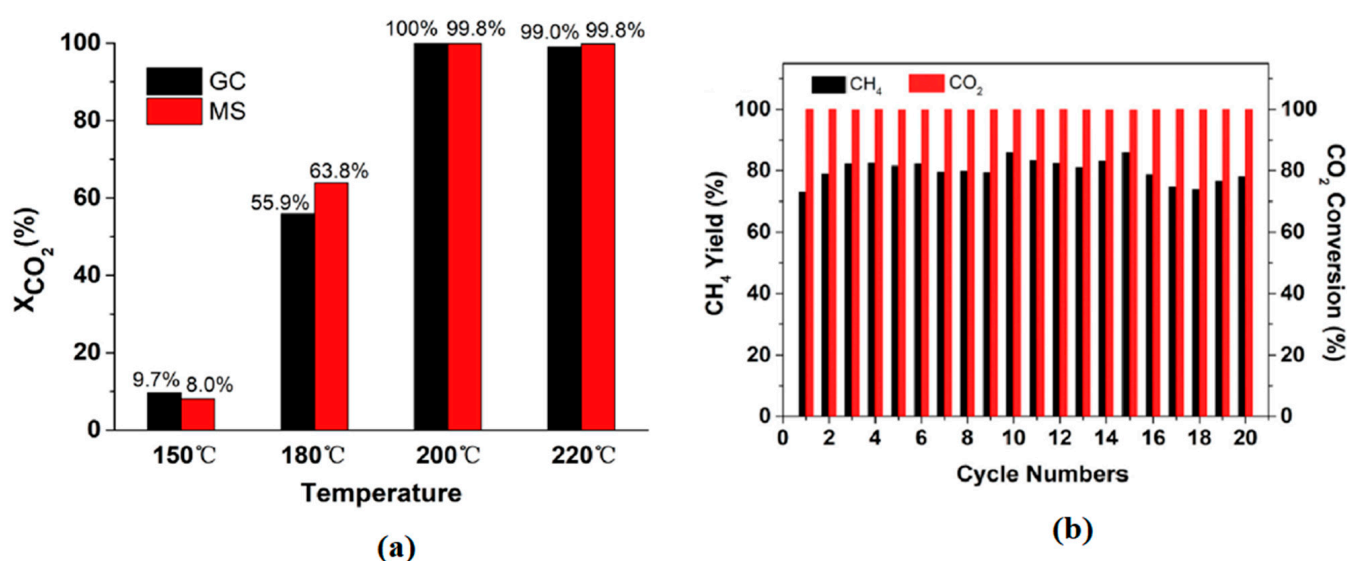


Figure 5. Catalytic activity of LaNi₅H₅ (a) CO₂ conversion at varying temperatures on a 20 h time of stream measured by GC and MS, and (b) CO₂ conversion and methane yield during the 20-cycle stability test. Adapted from Zhong et al. [56].

MgH₂/Cu_xO composite with a defect-rich surface was found to exhibit a 20.7% CO₂ conversion at 350 °C when using a high 5:1 ratio of CO₂:H₂ with high hydrogen utilisation efficiency. Although the composite hydride promoted the CO₂ conversion, a very low CH₄ selectivity of 45% was observed and attributed to the high selectivity towards C₂=C₄ olefins. In contrast, the CH₄ selectivity increased to 90% upon decreasing the CO₂:H₂ ratio to 1:10. The catalytic activity was reported to be a result of defective rich oxygen vacant Cu_xO and MgH₂ surfaces which acted as active sites and facilitated the adsorption and activation of CO₂, and subsequent CO₂ hydrogenation by lattice H[−] [50]. Metal active phase can be necessary to enhance the activity of MgH₂ hydride as well as lowering the reaction temperature according to Le Chatelier's principle for CO₂ hydrogenation, while maintaining high CH₄ yield. Amica et al. [58] achieved a CH₄ yield of 78% over MgH₂-doped with 10 wt.% Co at 350 °C for 48 h and a 4:1 molar ratio of MgH₂:CO₂ as optimum reaction conditions. However, a 44.6% CH₄ yield was obtained on the MgH₂ without cobalt (Co) as a catalyst. A high CH₄ yield on the Co-doped catalyst was reported to be due to the presence of the Co, which favors the prevailing methanation mechanism responding to the Sabatier process. MgNiH₄ and Mg₂FeH₆ complex hydrides were also applied as hydrogen sources and promoters for the CO₂ hydrogenation to methane [59].

Both complex hydrides initiated and facilitated different reaction pathways (reverse water gas shift followed by CO methanation for $\text{Mg}_2\text{FeH}_6\text{-CO}_2$ in the presence of steam and the global Sabatier methanation process for $\text{MgNiH}_4\text{-CO}_2$). In addition, they achieved 100% CO_2 conversion at 400 °C after 10 and 5 h, respectively, with a high yield of CH_4 as was detected by the Fourier transform infrared (FTIR) spectrometer and the formation of MgO by powder X-ray diffraction (PXRD) spectroscopy. CO_2 methanation reaction was conducted using activated MgH_2 and CaH_2 alkaline earth metal hydrides [60]. Although, both metal hydrides exhibited better activity, the highest yield (88%) and productivity (89%) of CH_4 was achieved with CaH_2 at 450 °C after 48 h. This indicated that the productivity of CH_4 can be largely influenced by the choice of alkaline earth metal hydride, reaction time and reaction temperature during CO_2 hydrogenation reaction.

Ball-milled alkali metal hydrides also start to emerge in the CO_2 hydrogenation process. On one hand, the highest CH_4 mole percentages of 74.7 and 90.3% were reported for LiH and NaH , respectively, in the hydrogenation of CO_2 over alkali metal hydrides (Figure 6). However, LiH achieved the maximum CH_4 yield of 80.15%, meanwhile 57.3% CH_4 yield was observed for NaH [63]. This indicated that the yield and mole % of CH_4 in both systems depended on the reaction temperature and time. On the other hand, Zhao et al. [64] studied the CO_2 hydrogenation using activated LiAlH_4 and NaAlH_4 alkali metal aluminium hydrides. Using the same reaction conditions, the maximum yield of 55% CH_4 was reached at 450 °C with NaAlH_4 , in comparison to 45% over LiAlH_4 . This is elucidated in Figure 7.

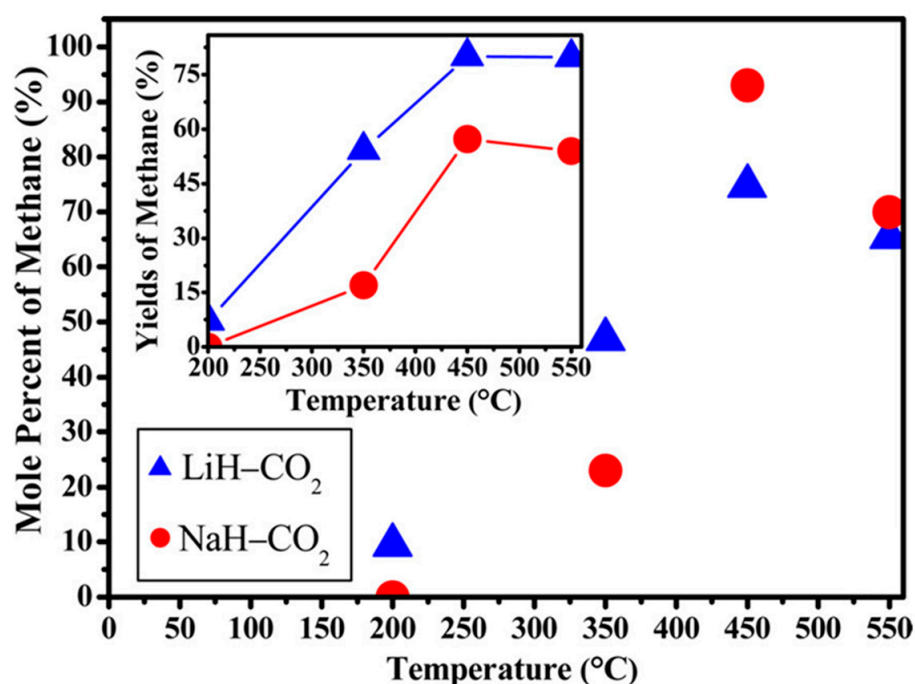


Figure 6. Reaction yields and mole percent of methane for the ball-milled LiH and NaH with CO_2 [$\text{MH}/\text{CO}_2 = 4:1$ mol/mol and 0.25 MPa, MH (M = Li or Na)] at range of temperatures; 200, 350, 450, and 550 °C for TOS of 48 h. Adapted from Dong et al. [63].

2.2.2. Synthesis of Formic Acid and Methanol over Metal Hydrides

Nilwanna et al. [65] used density functional theory (DFT) to study the reaction mechanism of aluminium-based metal–organic framework (Al-MOF) supported metal (II) hydrides (metal = Fe, Ni, Mn, Cu and Co) on the hydrogenation of CO_2 to formic acid. Although all the 5 Al-MOF supported metal (II) hydrides (MH-Al-MOF) have proven hydrogenation of formate to formic acid to be the rate-determining step, the most active

catalyst for hydrogenation of CO₂ to formic acid was the Al-MOF supported Ni hydride. Note that biphenyl-4,4'-dicarboxylate (DUT-5) is used as one of Al-MOFs. Refer to Table 4 for rate constants for the rate-determining step and relative Gibbs free energies for the CO₂ hydrogenation to formic acid on biphenyl-4,4'-dicarboxylate aluminium-based metal–organic framework supported metal hydrides (MH-DUT-5).

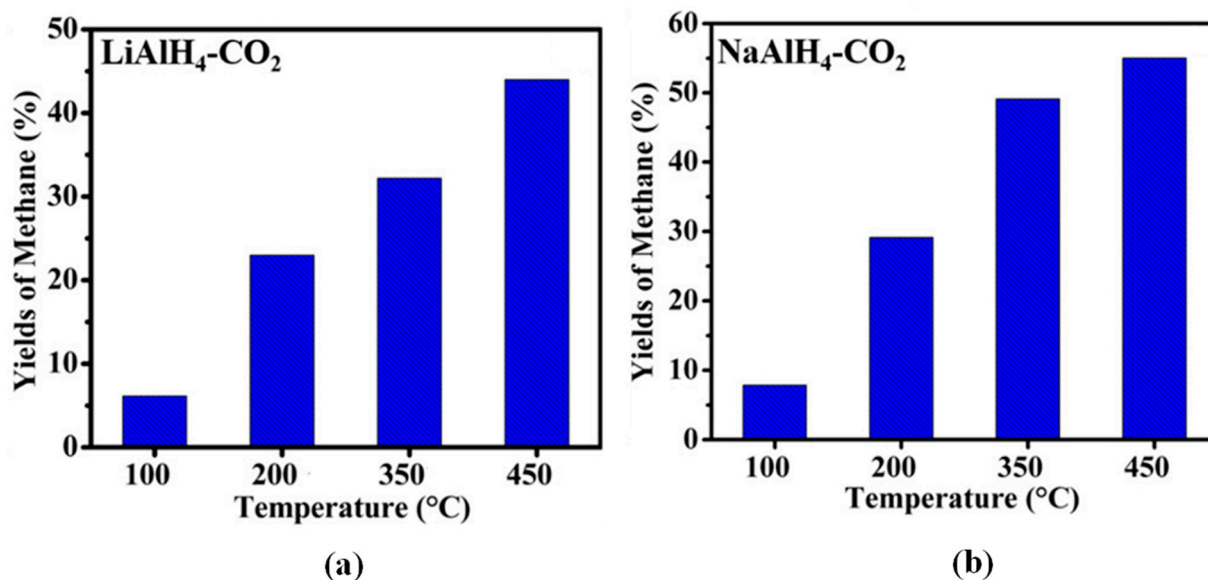


Figure 7. Yield of methane produced from the reactions between the ball-milled (a) LiAlH₄, and (b) NaAlH₄ with CO₂ [AAIH₄/CO₂ = 1/1 molar ratio and T = 48 h]. Adapted from Zhao et al. [64].

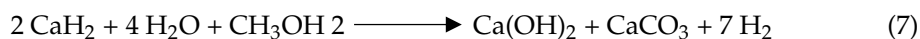
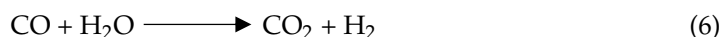
Table 4. Rate constants for the rate-determining step and relative Gibbs free energies for the CO₂ hydrogenation to formic acid on MH-DUT-5 (M = Mn, Fe, Co, Ni, and Cu).

MH-DUT-5	k (s ^{−1})	G° _{Ads} (kcal/mol)	G° _{Prod} (kcal/mol)
MnH-DUT-5	5.82 × 10 ^{−3}	−0.8	−3.3
FeH-DUT-5	5.89 × 10 ^{−2}	1.4	−2.0
CoH-DUT-5	7.58 × 10 ^{−7}	0.9	−3.3
NiH-DUT-5	5.29 × 10 ¹	−1.1	−6.7
CuH-DUT-5	1.08 × 10 ^{−3}	0.7	2.5

Where k = rate constant, G°_{Ads} = relative Gibbs free energy of adsorption, and G°_{Prod} = relative free energy of product.

Zhang et al. [66] studied the activation and hydrogenation of CO₂ over PtHn[−] clusters, and the formation of intermediates such as HPtCO₂[−] and H₂Pt(HCO₂)[−] was observed, suggesting the formate reaction pathway is energetically favorable. This can favor and promote the formic acid and/or methanol synthesis with subsequent hydrogenation of the formate species. A similar study by DFT calculations indicated that metal hydride MH (M = Cu, Ag, and Au) can initially lead to the formation of formate HCO₂[−] with AgH[−] as the most effective metal hydride anion followed by CuH[−] [67]. Since the methanol steam reforming (MSR) is a reverse reaction from the CO₂-hydrogenated methanol synthesis, it is paramount to study and understand the catalyst that can actively convert methanol and selectively produce H₂ and CO₂. In addition, it has to exhibit a better CO₂ absorption ability. Kadeer et al. [68] coupled the MSR with the hydrolysis of metal hydride. The employed CaH₂-CuO composite achieved a high hydrogen density of 6.5 wt. % (with 99% purity) at 250 °C. This is achieved by shifting the water gas shift reaction equilibrium towards the production of H₂ and CO₂ (Equation (6)) wherein the produced CO₂ from MSR could be absorbed by the hydroxide produced from hydrolysis of the metal hydride

(Equation (7)). This results in the production of highly purified H₂ and the suppression of CO formation. The high absorption capacity of the hydroxide component is an indication of the coupling effect of the MSR and hydrolysis of the metal hydride. In addition, the H₂ generated from the CaH₂-coupled MSR exhibited high H₂ density in comparison to conventional hydrolysis.



Kim et al. [69] studied the surface chemistry of CH₃OH on Pd(111)-H and Pd(111) to elucidate the effect of metal hydrides and C-O bond cleavage by decomposing CH₃OH as a function of temperature (300–750 K) using synchrotron-based ambient-pressure X-ray photoelectron spectroscopy (AP-XPS). They observed the surface and subsurface hydrides from Pd reacting with the decomposed CH_x and C/PdC_x species to produce a clean surface at 750 K. In contrast, a pristine Pd(111) led to the decomposition of the carbonaceous layer due to the lack of a hydride.

Pd is one of the most studied metal hydrides attributed to its high hydrogen storage capabilities by reversibly absorbing and desorbing H₂ under favorable conditions [70]. They can also act as a hydride of AB-type intermetallic alloy when doped with other transitional metals such as Cu, Ir, Ag, and Pt to form Pd-Cu-H, Pd-Ir-H, Pd-Ag-H, and Pd-Pt-H, respectively [71]. The reaction pathways were theoretically calculated for the hydrogenation of CO₂. Based on the energy barriers of the rate-limiting steps, it was clear that CH₃OH can be selectively produced through the formate pathway on the pristine MgH₂(001). On the other hand, the cluster Cu₇-MgH₂(001) favored the CH₄ through the RWGS pathway [72].

2.3. Effect of H₂ Dissociation and Formation of Metal Hydrides over Catalysts During CO₂ Hydrogenation to C1 Chemicals

During a catalytic reduction reaction, both CO₂ and H₂ molecule must undergo adsorption, activation and conversion on the surface of a heterogeneous catalyst. In addition to the study of CO₂ dissociation, which has been thoroughly reported [26,27], it is significant to understand the dissociation of the H₂ and how it affects the catalytic activity and selectivity during a catalytic reaction. Furthermore, it also gives insights in the formation and effect of in situ generated metal hydrides on catalysis. This will aid in selecting the best catalyst for the CO₂ reduction reaction. The cleavage of H₂ over a metal nanoparticle is broadly homolytic dissociative without an energy barrier. The low activation energy for the homolytic dissociation of H₂ on highly catalytic metals such as Ni, Cu, Pt and Pd, leads to low CO₂ hydrogenation selectivity due to the over-hydrogenation of CO₂ and its reaction intermediates on the metallic sites [73,74]. On the other hand, heterolytic dissociation presents limitations over the metal oxide for subsequent hydrogenation reactions, attributed to insufficient hydridic hydrogen when multiple hydrogen atoms and electrons are transferred, as well as an increase in the possibility of eliminating hydridic hydrogen to hydroxyls as a result of hydridic hydrogen migrating along the lattice oxygen [75]. In addition, it has much higher activation energy when compared to the H₂ homolytic dissociation. This leads to slower kinetics, making H₂ dissociation the rate-determining step with the necessity to employ high H₂ pressure and reaction temperature during reduction reactions [76]. It should be noted that both homolytic and heterolytic dissociation pathways of H₂ can occur consecutively on the surface of a metal. However, this depends on the effect of coordination with the support or metal cation. The H₂ heterolytic dissociation pathway is favored for the hydrogenation of C=O, N=O, and C=N bonds due to the nature

and better capability of polar bonds to accept hydridic hydrogen and protons [77–79]. Due to the nature of metal cations to be separated by counter-anions on the surface or sub-surface of bulk metal oxides, they resemble the Lewis acid-base (acid = metal, base = oxide) pair criterion. Metal oxides heterolytically dissociates the H_2 molecule into a proton and a hydridic hydrogen, and can exhibit excellent catalytic performance in the hydrogenation of CO_2 [79]. The active sites responsible for the H_2 heterolytic dissociation are attributed to the defect sites formed on the surface of metal oxide as well as their Lewis acid-base property.

To have a better understanding of the interaction between H_2 and a metal oxide during a hydrogenation reaction, Wu et al. [80] employed in situ inelastic neutron scattering spectroscopy (INS) and evidenced the heterolytic dissociation of H_2 into Ce-H hydride and O-H hydroxyls over defective ceria, with induced vacant sites of oxygen. Furthermore, they evidenced surface OHs on an oxygen vacant free or perfect ceria through electron and hydride transfer (Figure 8). This highlighted that both hydrides and oxygen vacancies over ceria can be formed via H_2 dissociation during reduction. Zinc oxide (ZnO) can serve as an active phase, promoter or support in the synthesis of methanol during CO_2 hydrogenation; therefore, a review on the surface chemistry of ZnO was conducted. From the study, the Zn-H hydride bands were observed using infrared spectroscopy when the surface of ZnO was exposed to gaseous hydrogen gas [81]. Refer to Figure 9 for the H_2 dissociation into a proton-hydride pair over ZnO.

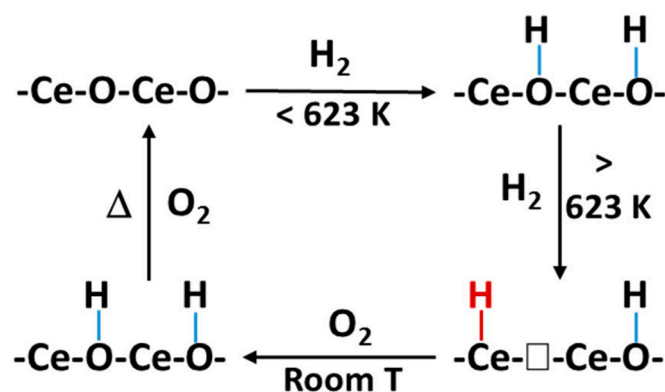


Figure 8. A schematic diagram of the H_2 interacting with CeO_2 forming surface Ce–H hydride and OH hydroxyl. Adapted from Wu et al. [80].

The dissociation of a polarised H_2 molecule into hydridic hydrogen (H^-) and proton (H^+) adsorbates on the surface of a catalyst can lead to better catalytic activity and selectivity for CO_2 hydrogenation. During the heterolytic H_2 dissociation over the surface of a defective or perfect Indium oxide (In_2O_3), the H^- binds to the In cation and the H^+ to the O anion. Similarly to the mechanism of CO_2 hydrogenation using ex situ synthesised metal hydrides, the hydridic hydrogen of the in situ generated metal hydride reacts with the C atom of the CO_2 adsorbed on the surface of the catalyst to activate and convert it into the desired C1 chemical such as methanol. On the other hand, the O in CO_2 reacts with the H^+ during protonation [81–83].

DFT calculations indicated that the oppositely charged Lewis acid-base pair sites of In and O atoms of the partially and perfectly reduced In_2O_3 can enhance the heterolytic H_2 dissociation. This promoted the hydride formation on In site which can be significant for methanol synthesis via the CO_2 hydrogenation [84]. Similarly, it was reported by Jia et al. [85] that oxygen vacant site and defects over the surface of In_2O_3 facilitate the heterolytic dissociation of H_2 , which can contribute to the CO_2 reduction with enhanced catalytic activity and selectivity. They clearly compare the H_2 activation and dissociation

on both stoichiometric and non-stoichiometric metal oxides, showing both the heterolytic and homolytic H_2 dissociation pathways on the surface of the defect and non-defect metal oxides. It can be noted that metal oxides have the compass to sequentially dissociate H_2 into both protons and proton-hydridic hydrogen pairs, absorb and react with CO_2 . As a result, the defect sites generated by high-temperature activation and the Lewis acid-base pair of the metal oxides serve as active sites for CO_2 catalytic hydrogenation reactions. Albeit, dissociative adsorption of H_2 was found to be facile via the heterolytic dissociation with moderate energy barrier on both $In_2O_3(111)$ and $In_2O_3(110)$ to form In-H and OH via DFT calculations, it was discovered that the diffusion of hydrogen facilitates homolytic adsorption positioning, with In_2O_3 surfaces hydroxylated by distinct types of OH groups. This resulted in changes in In oxidation states, suggesting that oxidation-reduction properties are also vital for the catalytic hydrogenation of CO_2 [86]. On the other hand, despite the protonation and reduction of CO_2 via the surface bicarbonate species being more energetically favorable (-0.78 eV adsorption energy) in comparison to hydridic hydrogen transfer ($+0.33$ eV adsorption energy) on the surface of $In_2O_3(110)$, it was indicated that the formate route by hydride transfer ($InH + CO_2$) was more favorable over the bicarbonate species route [87]. This will lead to an increment in the selectivity of formate and its subsequent products such as formic acid and methanol. Meanwhile, the subsequent steps in the bicarbonate species route are highly endothermic ($+1.07$ eV adsorption energy) and could react with surface hydroxyls with ease and exhibit high H_2O and CO_2 selectivities. Juárez et al. [88] conducted an in situ IR and INS spectroscopic study of H_2 species present over Au/CeO_2 and evidenced the formation of Au-H hydride and bridging hydroxyl group on the supported catalyst upon exposure to H_2 gas. Furthermore, they observed that bridging hydroxyl group bands at 400 to 650 cm^{-1} and 3690 cm^{-1} , and hydride band at 2126 cm^{-1} disappeared upon subsequent treatment with CO_2 . The disappearance of hydrogen species indicates consumption and catalytic activity with CO_2 . However, it was not possible to observe the Au-H band in INS due to the low percentage of Au ($0.48\text{ wt. \%}$) on the supported catalyst, and lower sensitivity of the INS.

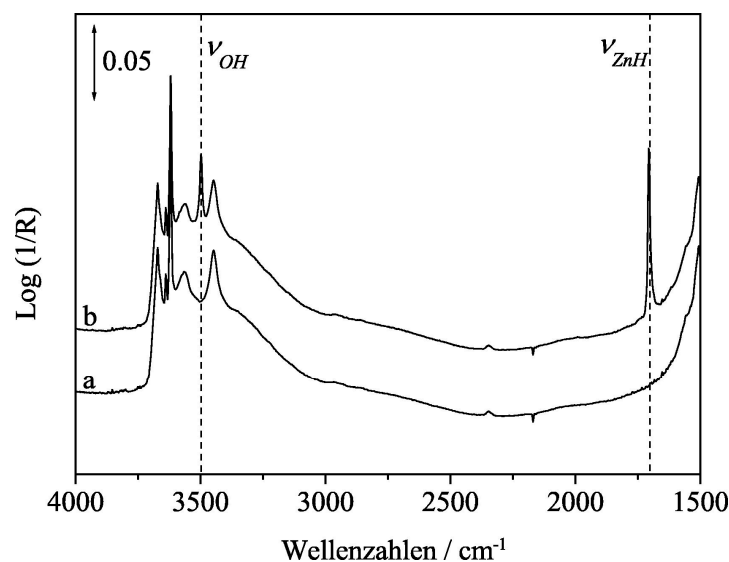


Figure 9. IR spectra of ZnO (a) exposed to He (b) Zn—H hydride formed upon H_2 exposure. Adapted from Strunk et al. [81].

In order to promote energy efficiency during CO_2 hydrogenation, Mao et al. [89] successfully employed an alkaline earth metal (Ca) to selectively activate and hydrogenate solid CO_2 captured by CaO in the form of $CaCO_3$ using ball milling at room temperature. A methane yield and selectivity of 56.1 and 100% were achieved, respectively. Furthermore,

the activity was increased upon doping the Ca-CaCO₃-H₂ with 9 wt. % Ni to 59.05% CH₄ yield and 95.35% CaCO₃ conversion.

2.4. Oxyhydrides for the CO₂ Hydrogenation

Attributed to their abilities to act as efficient active centers and/or supports, metal oxyhydride can also serve as catalysts for energy- and environmental-related catalytic reactions. Oxyhydrides are also starting to be explored in the CO₂ reduction. The catalytic hydrogenation of CO₂ into CH₄ was enhanced when BaTiO_{2.4}H_{0.6} and BaTiO_{2.67}H_{0.33} oxyhydrides were used as a support for Ni and Ru, respectively, in comparison to the Ru/BaTiO₃ and Ni/BaTiO₃ supported perovskite catalysts [90]. The catalytic activity increased by 2–7 folds over the perovskite-type oxyhydrides supported Ni and Ru (Figure 10) at 350 °C and 1 atm, under the 90 mL.min^{−1} flow-rate (H₂:CO₂ = 4:1) optimum reaction conditions. The improved activity was attributed to the reduction in H₂ poisoning, possibly influenced by the presence of hydridic hydrogen on oxyhydride (TiO_{2.4}H_{0.6} and BaTiO_{2.67}H_{0.33}) support. Moreover, Ru/BaTiO₃ has been reported to suffer from H₂ poisoning. The space-time yield and selectivity of methanol over Cu/BaTiO₃ increased significantly from 50 to 146 mg.g^{−1}cat.h^{−1} and 48 to 68%, respectively (Figure 11), upon partially replacing the O^{2−} anion of the BaTiO₃ perovskite with hydride to Cu/BaTiO_{2.8}H_{0.2} [91]. This was attributed to their ability to increase the local electronic density of the active phase on the metal support interface, which can lead to better promotion of CO₂ activation and conversion. Based on recent reported studies, it is clear that metal oxyhydrides are starting to receive attention in the CO₂ hydrogenation to C1 value-added products [90,91]. As a result, this can assist in mitigating the ever-rising emissions of CO₂ and sustain the energy supply by selectively producing synthetic gas CH₄ which can be used as an energy carrier or directly as a renewable fuel. However, not much has been reported on CO₂ hydrogenation to formic acid and methanol using metal hydrides. Therefore, a good opportunity is presented to further explore the chemistry of metal hydride heterogeneous catalysts on the synthesis of other C1 chemicals. The next chapter briefly highlights carbon dioxide reduction over homogeneous metal hydride catalysts. This is to elucidate the catalytic importance of hydrides on methanol synthesis via CO₂ hydrogenation.

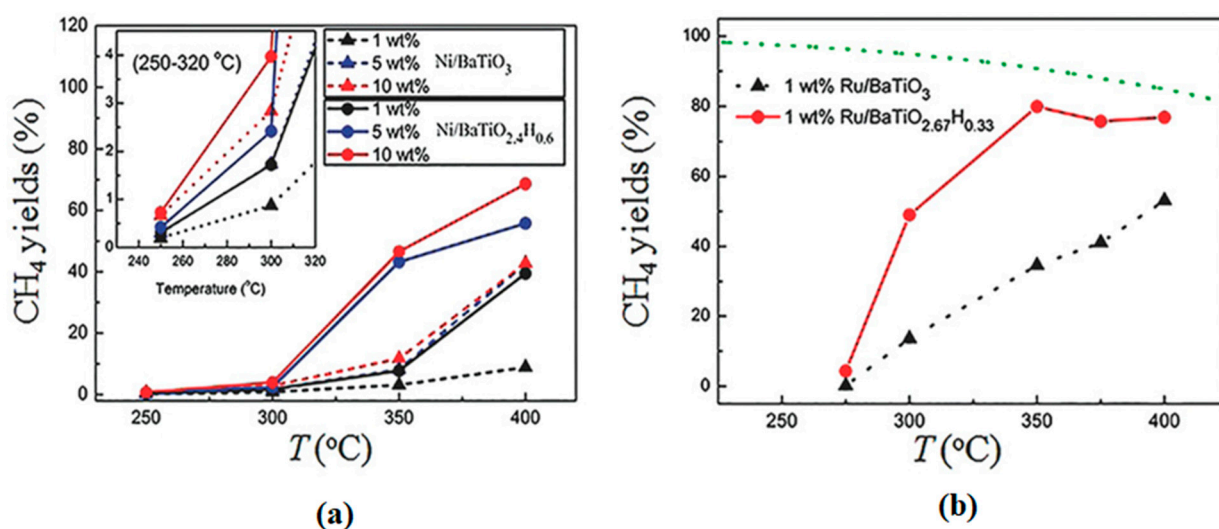


Figure 10. Cont.

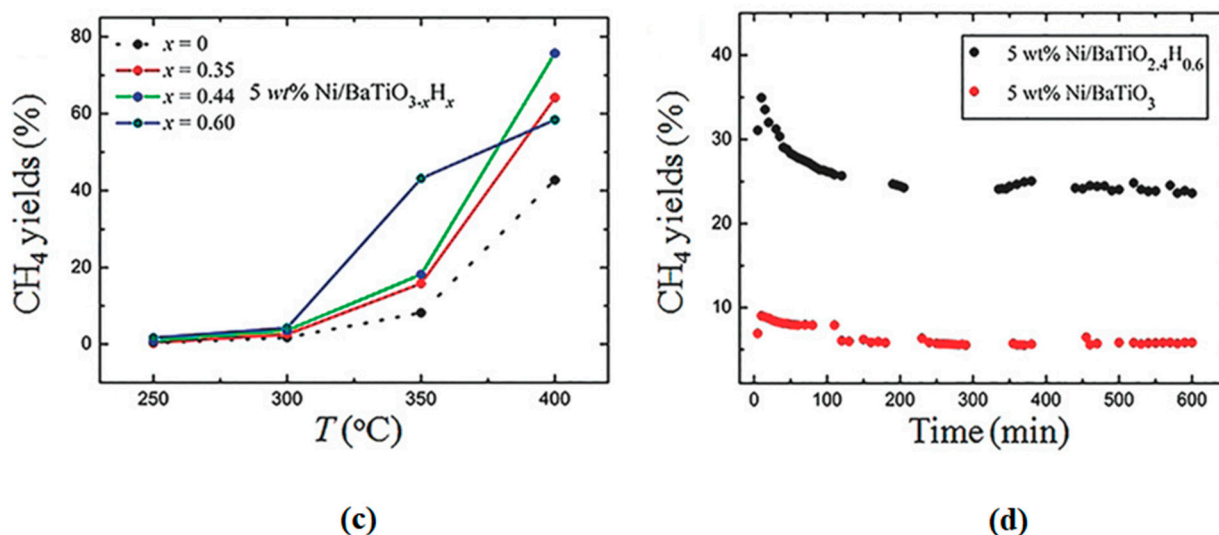


Figure 10. CH₄ yield at different reaction temperatures of (a) Ni and (b) Ru supported pristine perovskite and oxyhydride catalysts. (c) CH₄ yield over various contents (x) of Ni/BaTiO_{3-x}H_x and (d) CH₄ yield over 10 h stability test. Adapted from Tang et al. [90].

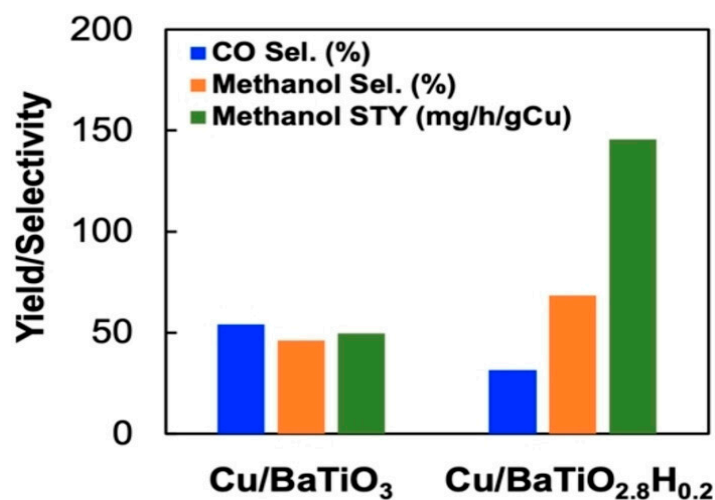


Figure 11. CO₂ hydrogenation over Cu/BaTiO₃ and Cu/BaTiO_{2.8}H_{0.2}. Adapted from He et al. [91].

3. Hydrogenation of CO₂ over Homogeneous Transition Metal Hydrides Catalysts

In comparison to CO₂ hydrogenation by late (groups 8–10) transition metal complexes which suffer from the production of 2e[−] reduction products such as formate and CO, a review was conducted for the CO₂ reduction with early (groups 3–7) transition metal complexes to form formaldehyde, methoxide species and their subsequent hydrogenated products [92]. Methanol can be observed in the stoichiometric and catalytic conversion attributed to the ability of groups 3–7 metal complexes to pass electrons and the Lewis acidity (of the early metals) to activate CO₂ (Figure 12). On the other hand, earth-abundant 3d transition metals have shown significant increases in catalytic activity in comparison to second (4d) and third (5d) row transition metal complexes in CO₂ reduction into formic acid and methanol [93]. The conversion of CO₂ into methanol via hydrogenation with renewable hydrogen over homogeneous metal hydrides is an emerging field of research. This is due to the high activity of homogeneous metal hydride catalysts. The chemistry of hydride transfers from transition metal complex families and applications on CO₂ reduction to fuels and chemicals such as; methanol and formates equivalents have been well

documented [93–96]. Known hydricity values and free energy of the electron reduction ($E_{1/2}(M^{n+/(n-1)+})$) of the parent complex from the sets of data consisting of 51 transition metal hydride complexes were established [41]. From the data sets, a linear trend was observed which promotes hydricity estimation on the parent complex's reduction potential and guides the design and optimisation of the metal hydride complex for CO₂ hydrogenation to lower C-chemicals. Although transition metal complexes and their respective hydrides have a better CO₂ catalytic activity towards methanol, transition metal complexes suffer from poor product and catalyst separation, and purification. This leads to their limitation to be applied at a large industrial scale.

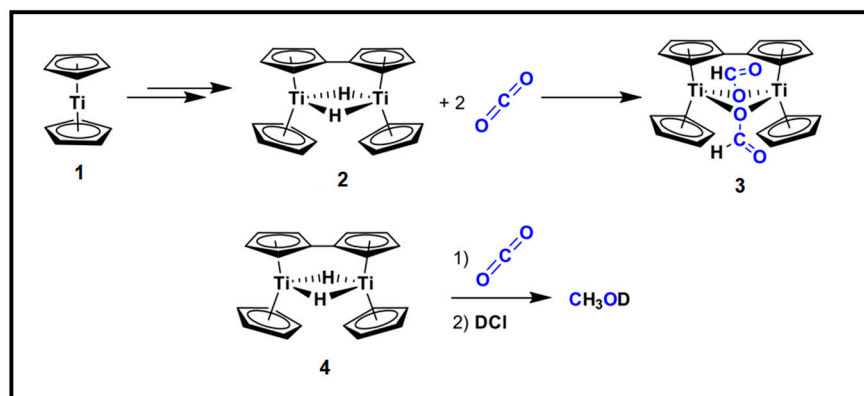


Figure 12. Schematic representation of the methanol synthesis from CO₂ over group 4's titanium hydride complex. Adapted from Grice. [92].

4. Conclusions

Continuous utilisation of fossil fuels has led to an accelerated rate of global warming beyond the pre-industrial stage. This is due to the excessive emissions of anthropogenic CO₂ released into the environment. This review presents how CO₂ can be converted into C1 chemicals namely, CH₄, HCOOH, and CH₃OH using renewable green hydrogen. The CO₂ hydrogenation conversion can mitigate CO₂ emissions, maintain energy security, and replace depleting fossil resources. Catalysts are applied and play a very important role in CO₂ hydrogenation into C1 chemicals. The produced C1 chemicals can be directly used as substitute fuels and carriers or storage for clean energy. Metal hydrides are starting to emerge in the thermo-catalytic hydrogenation of CO₂. This is attributed to their enhanced hydrogen adsorption–desorption reversibility, and newly developed active and abundant sites. They can be classified into five different categories based on the type of metal–hydrogen bonding namely, ionic, interstitial, covalent, complex, and high-pressure metal hydrides. Each of these classes have benefits and drawbacks. Various types of complex and ionic metal hydrides applied in the CO₂ methanation are reviewed. Wherein, a high CO₂ conversion and CH₄ selectivity are reported. This is attributed to their high reductibility by hydridic hydrogen (H[−]) during hydrogenation reactions at different temperatures, pressures, and space velocities. The hydridic hydrogen can activate and convert CO₂ through the reverse water gas shift (RWGS) followed by CO methanation and the global Sabatier reactions. The introduction of transition, such as Co and Ni to the metal hydrides, enhances the CH₄ selectivity. Furthermore, alkali and alkaline earth metals can act as promoters and hydridic hydrogen sources during the CO₂ methanation.

Although, metal hydrides have been thoroughly reported regarding CO₂ methanation, few studies have been conducted on the CO₂ hydrogenation to other C1 (CH₃OH and HCOOH) chemicals. We review the application of metal hydrides on the non-catalytic process for pure H₂ and CO₂ production from CH₃OH. This highlights the importance and role of metal hydrides on the reverse pathway of CO₂ hydrogenation to CH₃OH.

The application of metal hydrides on the methanol steam reforming and decomposition reactions enhanced the production and selectivity of both H_2 and CO_2 , suppressing the RWGS products ($CO + H_2O$).

Hydridic hydrogen formed in situ during catalytic hydrogenation reaction is also reviewed. Both homolytic and heterolytic H_2 dissociation can occur simultaneously during hydrogenation. This was observed from spectroscopy experiments. However, the H_2 heterolytic dissociation into proton (H^+) and hydridic hydrogen (H^-) is favoured for the hydrogenation of $C=O$, $N=O$, and $C=N$ polar bonds. Metal oxides heterolytically dissociate H_2 into H^+ and H^- and positively influence the catalytic performance. DFT calculations also suggest that heterolytic dissociation on metal oxide promotes hydridic hydrogen-proton pairs. The hydridic hydride binds to the metal cation, while the proton binds to the oxygen anion ($M-H$ and $O-H$).

Analogous to metal hydrides, metal oxyhydrides are also starting to find a way into CO_2 hydrogenation. Perovskite-type oxyhydride-supported transition metals enhanced the selectivity and space time yield of methanol.

We further elucidated the catalytic importance of metal hydrides on the synthesis of C1 chemicals—mainly CH_3OH —by briefly introducing homogeneous transition metal hydrides. Significant increases in catalytic activity and CH_3OH selectivity are reported due to the presence of a 3d and group 3–7 metal complexes ability to surpass $2 e^-$ reduction of CO_2 . Although homogeneous complex metal hydrides have a better catalytic activity, they suffer from poor product and catalyst separation. Therefore, their application is limited at a large scale in industry.

5. Future Outlook and Perspective

The thermo-catalytic hydrogenation of CO_2 into fuels and chemicals has been studied for many years and has a substantial technical flexibility. Significant progress has been made in CO_2 hydrogenation to C1 chemicals over traditional heterogeneous catalysts. However, metal hydrides are only starting to emerge in the area of CO_2 hydrogenation. As of yet, there is very limited research on the formic acid and methanol synthesis over metal hydrides. The emergence of advanced in situ characterisation techniques and artificial intelligence (AI) aided probing or predictions, is anticipated to promote discoveries in metal hydride catalysed reactions. In this case, the nature of chemical bonding during the heterolytic dissociation of H_2 , hydride transfer and phase transitions, and changes in local electron density of metal hydrides that significantly influence catalytic performance (activity, selectivity and stability) will be well-understood. Moreover, the different reaction pathways over metal hydrides that give a proper understanding of the effect of the hydrides in selectively producing desired products of interest will be unravelled. Therefore, it is anticipated that the rational design and proper synthesis of supported or unsupported metal hydride catalysts will aid in the adsorption, activation and conversion of the thermodynamically stable and kinetically inert CO_2 at a much faster reaction rate, as well as producing highly selective methane, formic acid and methanol products.

Author Contributions: Conceptualization, F.C., N.M.M. and X.O.; writing—original draft preparation, M.A.M.; writing—review and editing, M.A.M., X.O., N.M.M. and F.C.; supervision, F.C., N.M.M. and X.O.; funding acquisition, F.C. All authors have read and agreed to the published version of the manuscript.

Funding: This research was supported by Zhejiang Provincial Natural Science Foundation of China under grant No. LR24B030002.

Data Availability Statement: Data are available on request.

Acknowledgments: The authors gratefully acknowledge the Yongjiang Laboratory and the Nottingham Ningbo China Beacons of Excellence Research and Innovation Institute, and the faculty of science and engineering of the University of Nottingham Ningbo China for the administrative support in writing this review.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

MH	Metal hydrides
GHG	Greenhouse gas
INS	Inelastic neutron scattering spectroscopy
TOS	Time on stream
WHSV	Weight hourly space velocity
AP-XPS	Ambient-pressure X-ray photoelectron spectroscopy
XRD	X-ray spectroscopy
IR	Infrared spectroscopy
DFT	Density functional theory

References

1. Khan, A.A.; Tahir, M. Recent advancements in engineering approach towards design of photo-reactors for selective photocatalytic CO₂ reduction to renewable fuels. *J. CO₂ Util.* **2019**, *29*, 205–239. [\[CrossRef\]](#)
2. Zhong, J.; Yang, X.; Wu, Z.; Liang, B.; Huang, Y.; Zhang, T. State of the art and perspectives in heterogeneous catalysis of CO₂ hydrogenation to methanol. *Chem. Soc. Rev.* **2020**, *49*, 1385–1413. [\[CrossRef\]](#)
3. Wei, L.; Yu, C.; Zhang, Q.; Liu, H.; Wang, Y. TiO₂-based heterojunction photocatalysts for photocatalytic reduction of CO₂ into solar fuels. *J. Mater. Chem. A* **2018**, *6*, 22411–22436. [\[CrossRef\]](#)
4. Li, K.; Teng, C.; Wang, S.; Min, Q. Recent Advances in TiO₂-Based Heterojunctions for Photocatalytic CO₂ Reduction with Water Oxidation: A Review. *Front. Chem.* **2021**, *9*, 637501. [\[CrossRef\]](#)
5. Setia, D.; Roithová, J. Trends in the hydricities of iron, cobalt, and nickel complexes and the metal-hydride reactivities with CO₂. *Int. J. Mass Spectrom.* **2024**, *504*, 117310. [\[CrossRef\]](#)
6. Younas, M.; Loong Kong, L.; Bashir, M.J.K.; Nadeem, H.; Shehzad, A.; Sethupathi, S. Recent Advancements, Fundamental Challenges, and Opportunities in Catalytic Methanation of CO₂. *Energy Fuels* **2016**, *30*, 8815–8831. [\[CrossRef\]](#)
7. Frei, M.S.; Mondelli, C.; García-Muelas, R.; Kley, K.S.; Puértolas, B.; López, N.; Safonova, O.V.; Stewart, J.A.; Ferré, D.C.; Pérez-Ramírez, J. Atomic-scale engineering of indium oxide promotion by palladium for methanol production via CO₂ hydrogenation. *Nat. Commun.* **2019**, *10*, 3377. [\[CrossRef\]](#)
8. Lv, X.; Lu, G.; Wang, Z.Q.; Xu, Z.N.; Guo, G.C. Computational Evidence for Lewis Base-Promoted CO₂ Hydrogenation to Formic Acid on Gold Surfaces. *ACS Catal.* **2017**, *7*, 4519–4526. [\[CrossRef\]](#)
9. Reutemann, W.; Kieczka, H. Formic acid. *Ullmann's Encycl. Ind. Chem.* **2012**, *16*, 13–31. [\[CrossRef\]](#)
10. Pérez-Fortes, M.; Schöneberger, J.C.; Boulamanti, A.; Tzimas, E. Methanol synthesis using captured CO₂ as raw material: Techno-economic and environmental assessment. *Appl. Energy* **2016**, *161*, 718–732. [\[CrossRef\]](#)
11. Cordero-Lanzac, T.; Ramirez, A.; Navajas, A.; Gevers, L.; Brunialti, S.; Gandía, L.M.; Aguayo, A.T.; Sarathy, S.M.; Gascon, J. A techno-economic and life cycle assessment for the production of green methanol from CO₂: Catalyst and process bottlenecks. *J. Energy Chem.* **2022**, *68*, 255–266. [\[CrossRef\]](#)
12. Iribarren, D.; Calvo-Serrano, R.; Martín-Gamboa, M.; Galán-Martín, A.; Guillén-Gosálbez, G. Social life cycle assessment of green methanol and benchmarking against conventional fossil methanol. *Sci. Total Environ.* **2022**, *824*, 153840. [\[CrossRef\]](#)
13. Zhang, X.; Zhang, G.; Song, C.; Guo, X. Catalytic Conversion of Carbon Dioxide to Methanol: Current Status and Future Perspective. *Front. Energy Res.* **2021**, *8*, 621119. [\[CrossRef\]](#)
14. Cheng, D.; Negreiros, F.R.; Aprà, E.; Fortunelli, A. Computational approaches to the chemical conversion of carbon dioxide. *ChemSusChem* **2013**, *6*, 944–965. [\[CrossRef\]](#)
15. Lykaki, M.; Mandela, E.; Varvoutis, G.; Lampropoulos, A.; Marnellos, G.E.; Konsolakis, M. State-of-the-art thermocatalytic systems for CH₄ and CO production via CO₂ hydrogenation: Critical comparison, mechanistic considerations and structure-performance insights. *Discov. Chem. Eng.* **2024**, *4*, 11. [\[CrossRef\]](#)
16. Wang, M.; Luo, L.; Wang, C.; Du, J.; Li, H.; Zeng, J. Heterogeneous Catalysts toward CO₂ Hydrogenation for Sustainable Carbon Cycle. *Accounts Mater. Res.* **2022**, *3*, 565–571. [\[CrossRef\]](#)

17. Álvarez, A.; Bansode, A.; Urakawa, A.; Bavykina, A.V.; Wezendonk, T.A.; Makkee, M.; Gascon, J.; Kapteijn, F. Challenges in the Greener Production of Formates/Formic Acid, Methanol, and DME by Heterogeneously Catalyzed CO₂ Hydrogenation Processes. *Chem. Rev.* **2017**, *117*, 9804–9838. [\[CrossRef\]](#)
18. Jacquemin, M.; Beuls, A.; Ruiz, P. Catalytic production of methane from CO₂ and H₂ at low temperature: Insight on the reaction mechanism. *Catal. Today* **2010**, *157*, 462–466. [\[CrossRef\]](#)
19. Maru, M.S.; Ram, S.; Shukla, R.S.; Khan, N.H. Ruthenium-hydrotalcite (Ru-HT) as an effective heterogeneous catalyst for the selective hydrogenation of CO₂ to formic acid. *Mol. Catal.* **2018**, *446*, 23–30. [\[CrossRef\]](#)
20. Shen, C.; Sun, K.; Zou, R.; Wu, Q.; Mei, D.; Liu, C.J. CO₂ Hydrogenation to Methanol on Indium Oxide-Supported Rhenium Catalysts: The Effects of Size. *ACS Catal.* **2022**, *12*, 12658–12669. [\[CrossRef\]](#)
21. Monene, T.J.; Cele, M.N. The activities of selected metals and supports for CO₂ hydrogenation to methanol: A review. *Appl. Catal. O Open* **2024**, *197*, 207019. [\[CrossRef\]](#)
22. Wang, J.; Zhang, G.; Zhu, J.; Zhang, X.; Ding, F.; Zhang, A.; Guo, X.; Song, C. CO₂ Hydrogenation to Methanol over In₂O₃-Based Catalysts: From Mechanism to Catalyst Development. *ACS Catal.* **2021**, *11*, 1406–1423. [\[CrossRef\]](#)
23. Grinter, D.C.; Graciani, J.; Palomino, R.M.; Xu, F.; Waluyo, I.; Sanz, J.F.; Senanayake, S.D.; Rodriguez, J.A. Adsorption and activation of CO₂ on Pt/CeO_x/TiO₂(110): Role of the Pt-CeO_x interface. *Surf. Sci.* **2021**, *710*, 121852. [\[CrossRef\]](#)
24. Chinchin, G.C.; Spencer, M.S.; Waugh, K.C.; Whan, D.A. Promotion of methanol synthesis and the water-gas shift reactions by adsorbed oxygen on supported copper catalysts. *J. Chem. Soc. Faraday Trans. 1* **1987**, *83*, 2193–2212. [\[CrossRef\]](#)
25. Eren, B.; Weatherup, R.S.; Liakakos, N.; Somorjai, G.A.; Salmeron, M. Dissociative Carbon Dioxide Adsorption and Morphological Changes on Cu(100) and Cu(111) at Ambient Pressures. *J. Am. Chem. Soc.* **2016**, *138*, 8207–8211. [\[CrossRef\]](#)
26. Su, H.; Ye, Y.; Lee, K.J.; Zeng, J.; Mun, B.S.; Crumlin, E.J. Probing the surface chemistry for reverse water gas shift reaction on Pt(111) using ambient pressure X-ray photoelectron spectroscopy. *J. Catal.* **2020**, *391*, 123–131. [\[CrossRef\]](#)
27. Sheffield, J.; Martin, K.B. *Alternative Fuels and Advanced Vehicle Technologies for Improved Environmental Performance*; Woodhead Publishing Ltd.: Sawston, UK, 2014; pp. 117–135.
28. Gay-Lussac, J.L.; Thenard, L.J. Recherches physico-chimiques. *Paris* **1811**, *2*, 1–57.
29. Hautefeuille, P.; Troost, L. Alliages de l'hydrogène avec les métaux alcalins. *Ann. Chim. Phys.* **1874**, *2*, 273.
30. Maeno, Z.; Yasumura, S.; Wu, X.; Huang, M.; Liu, C.; Toyao, T.; Shimizu, K. Isolated Indium Hydrides in CHA Zeolites: Speciation and Catalysis for Nonoxidative Dehydrogenation of Ethane. *J. Am. Chem. Soc.* **2020**, *142*, 4820–4832. [\[CrossRef\]](#)
31. Yu, H.; Li, X.; Zheng, J. Beyond Hydrogen Storage: Metal Hydrides for Catalysis. *ACS Catal.* **2024**, *14*, 3139–3157. [\[CrossRef\]](#)
32. Wang, Q.; Guo, J.; Chen, P. Complex transition metal hydrides for heterogeneous catalysis. *Chem Catal.* **2023**, *3*, 100524. [\[CrossRef\]](#)
33. Schneemann, A.; White, J.L.; Kang, S.Y.; Jeong, S.; Wan, L.F.; Cho, E.S.; Heo, T.W.; Prendergast, D.; Urban, J.J.; Wood, B.C.; et al. Nanostructured Metal Hydrides for Hydrogen Storage. *Chem. Rev.* **2018**, *118*, 10775. [\[CrossRef\]](#) [\[PubMed\]](#)
34. Wietelmann, U.; Felderhoff, M.; Rittmeyer, P. Hydrides. In *Ullmann's Encyclopedia of Industrial Chemistry*; Wiley: Berlin, Germany, 2016; pp. 1–36.
35. Luo, Y.; Wang, Q.; Li, J.; Xu, F.; Sun, L.; Zou, Y.; Chu, H.; Li, B.; Zhang, K. Enhanced hydrogen storage/sensing of metal hydrides by nanomodification. *Mater. Today Nano* **2020**, *9*, 100071. [\[CrossRef\]](#)
36. Dekura, S.; Kobayashi, H.; Kusada, K.; Kitagawa, H. Hydrogen in Palladium and Storage Properties of Related Nanomaterials: Size, Shape, Alloying, and Metal–Organic Framework Coating Effects. *Phys. Chem. Chem. Phys.* **2019**, *20*, 1158–1176. [\[CrossRef\]](#)
37. Copéret, C. C–H bond activation and organometallic intermediates on isolated metal centers on oxide surfaces. *Chem. Rev.* **2010**, *110*, 656–680. [\[CrossRef\]](#) [\[PubMed\]](#)
38. Copéret, C.; Estes, D.P.; Larmier, K.; Searles, K. Isolated Surface Hydrides: Formation, Structure, and Reactivity. *Chem. Rev.* **2016**, *116*, 8463–8505. [\[CrossRef\]](#)
39. Serykh, A.I. Low-Dimensional Indium Oxo-Species on the Surface of In₂O₃/Al₂O₃ Catalytic Material: The Sites of Dissociative Adsorption of Hydrogen. *J. Phys. Chem. C* **2016**, *120*, 21436–21440. [\[CrossRef\]](#)
40. Miyazaki, M.; Ogasawara, K.; Nakao, T.; Sasase, M.; Kitano, M.; Hosono, H. Hexagonal BaTiO_(3-x)H_x Oxyhydride as a Water-Durable Catalyst Support for Chemoselective Hydrogenation. *J. Am. Chem. Soc.* **2022**, *144*, 6453–6464. [\[CrossRef\]](#)
41. Waldie, K.M.; Ostericher, A.L.; Reineke, M.H.; Sasayama, A.F.; Kubiak, C.P. Hydricity of Transition-Metal Hydrides: Thermodynamic Considerations for CO₂ Reduction. *ACS Catal.* **2018**, *8*, 1313–1324. [\[CrossRef\]](#)
42. Niu, Y.; Borsodi, J.; Wootsch, A.; Révay, Z.; Hävecker, M.; Knop-Gericke, A.; Jackson, S.D.; Schlögl, R. Visualizing Formation of Intermetallic PdZn in a Palladium/Zinc Oxide Catalyst: Interfacial Fertilization by PdH_x. *Angew. Chemie—Int. Ed.* **2019**, *58*, 4232–4237. [\[CrossRef\]](#) [\[PubMed\]](#)
43. Teschner, D.; Borsodi, J.; Wootsch, A.; Révay, Z.; Hävecker, M.; Knop-Gericke, A.; Jackson, S.D.; Schlögl, R. The Roles of Subsurface Carbon and Hydrogen in Palladium-Catalyzed Alkyne Hydrogenation. *Science* **2008**, *320*, 86–89. [\[CrossRef\]](#)
44. Fovanna, T.; Nachtegaal, M.; Clark, A.H.; Kröcher, O.; Ferri, D. Preparation, Quantification, and Reaction of Pd Hydrides on Pd/Al₂O₃ in Liquid Environment. *ACS Catal.* **2023**, *13*, 3323–3332. [\[CrossRef\]](#)

45. Chang, F.; Tezsevin, I.; de Rijk, J.W.; Meeldijk, J.D.; Hofmann, J.P.; Er, S.; Ngene, P.; de Jongh, P.E. Potassium hydride-intercalated graphite as an efficient heterogeneous catalyst for ammonia synthesis. *Nat. Catal.* **2022**, *5*, 222–230. [[CrossRef](#)]
46. Chang, F.; Guan, Y.; Chang, X.; Guo, J.; Wang, P.; Gao, W.; Wu, G.; Zheng, J.; Li, X.; Chen, P. Alkali and Alkaline Earth Hydrides-Driven N₂ Activation and Transformation over Mn Nitride Catalyst. *J. Am. Chem. Soc.* **2018**, *140*, 14799–14806. [[CrossRef](#)]
47. Wang, P.; Chang, F.; Gao, W.; Guo, J.; Wu, G.; He, T.; Chen, P. Breaking scaling relations to achieve low-temperature ammonia synthesis through LiH-mediated nitrogen transfer and hydrogenation. *Nat. Chem.* **2017**, *9*, 64–70. [[CrossRef](#)] [[PubMed](#)]
48. Tang, Y.; Kobayashi, Y.; Masuda, N.; Uchida, Y.; Okamoto, H.; Kageyama, T.; Hosokawa, S.; Loyer, F.; Mitsuhara, K.; Yamanaka, K.; et al. Metal-Dependent Support Effects of Oxyhydride-Supported Ru, Fe, Co Catalysts for Ammonia Synthesis. *Adv. Energy Mater.* **2018**, *8*, 1801772. [[CrossRef](#)]
49. Kitano, M.; Inoue, Y.; Ishikawa, H.; Yamagata, K.; Nakao, T.; Tada, T.; Matsuiishi, S.; Yokoyama, T.; Hara, M.; Hosono, H. Essential role of hydride ion in ruthenium-based ammonia synthesis catalysts. *Chem. Sci.* **2016**, *7*, 4036–4043. [[CrossRef](#)]
50. Chen, H.; Liu, P.; Li, J.; Wang, Y.; She, C.; Liu, J.; Zhang, L.; Yang, Q.; Zhou, S.; Feng, X. MgH₂/Cu_xO Hydrogen Storage Composite with Defect-Rich Surfaces for Carbon Dioxide Hydrogenation. *ACS Appl. Mater. Interfaces* **2019**, *11*, 31009–31017. [[CrossRef](#)]
51. Feng, X.; Jiang, L.; Li, Z.; Wang, S.; Ye, J.; Wu, Y.; Yuan, B. Boosting the hydrogenation activity of dibenzyltoluene catalyzed by Mg-based metal hydrides. *Int. J. Hydrogen Energy* **2022**, *47*, 23994–24003. [[CrossRef](#)]
52. Lelis, M.; Varnagir, S.; Urbonavicius, M.; Zakarauskas, K. Investigation of catalyst development from Mg₂NiH₄ hydride and its application for the CO₂ methanation reaction. *Coatings* **2020**, *10*, 1178. [[CrossRef](#)]
53. Amica, G.; Gennari, F.C. Synergistic effect of MgH₂ doping with Ni and carbon nanotubes on thermochemical CO₂ recycling process for CH₄-H₂ mixtures production. *Int. J. Hydrogen Energy* **2022**, *47*, 428–442. [[CrossRef](#)]
54. Chen, H.; Liu, P.; Liu, J.; Feng, X.; Zhou, S. Mechanochemical in-situ incorporation of Ni on MgO/MgH₂ surface for the selective O-/C-terminal catalytic hydrogenation of CO₂ to CH₄. *J. Catal.* **2021**, *394*, 397–405. [[CrossRef](#)]
55. Grasso, M.L.; Puzskiel, J.; Gennari, F.C.; Santoru, A.; Dornheim, M.; Pistidda, C. CO₂ reactivity with Mg₂NiH₄ synthesized by in situ monitoring of mechanical milling. *Phys. Chem. Chem. Phys.* **2020**, *22*, 1944–1952. [[CrossRef](#)] [[PubMed](#)]
56. Zhong, D.; Ouyang, L.; Liu, J.; Wang, H. Metallic Ni nanocatalyst in situ formed from LaNi₅H₅ toward efficient CO₂ methanation. *Int. J. Hydrogen Energy* **2019**, *44*, 29068–29074. [[CrossRef](#)]
57. Chen, H.; Ma, N.; Cheng, C.; Zhang, H.; Yuan, W.; Liu, P. Hydrogen activation on aluminium-doped magnesium hydride surface for methanation of carbon dioxide. *Appl. Surf. Sci.* **2020**, *515*, 146038. [[CrossRef](#)]
58. Amica, G.; Azcona, R.; Aparicio, S.; Gennari, F.C. Catalysis effect on CO₂ methanation using MgH₂ as a portable hydrogen medium. *Phys. Chem. Chem. Phys.* **2020**, *22*, 14720–14730. [[CrossRef](#)]
59. Grasso, M.L.; Puzskiel, J.; Albanesi, F.; Dornheim, L.M.; Pistidda, C.; Gennari, F.C. CO₂ reutilization for methane production via a catalytic process promoted by hydrides. *Phys. Chem. Chem. Phys.* **2019**, *21*, 19825–19834. [[CrossRef](#)]
60. Zhao, J.; Wei, Y.F.; Cai, Y.L.; Wang, L.Z.; Xie, J.; Teng, Y.L.; Zhu, W.; Shen, M.; Dong, B.X. Highly Selective and Efficient Reduction of CO₂ to Methane by Activated Alkaline Earth Metal Hydrides without a Catalyst. *ACS Sustain. Chem. Eng.* **2019**, *7*, 4831–4841. [[CrossRef](#)]
61. Kato, S.; Matam, S.K.; Kerger, P.; Bernard, L.; Battaglia, C.; Vogel, D.; Rohwerder, M.; Zuttel, A. The Origin of the Catalytic Activity of a Metal Hydride in CO₂ Reduction. *Angew. Chemie—Int. Ed.* **2016**, *55*, 6028–6032. [[CrossRef](#)]
62. Kato, S.; Borgschulte, A.; Ferri, D.; Biemann, M.; Crivello, J.C.; Wiedenmann, D.; Parlinska-Wojtan, M.; Rossbach, P.; Lu, Y.; Remhof, A.; et al. CO₂ hydrogenation on a metal hydride surface. *Phys. Chem. Chem. Phys.* **2012**, *14*, 5518–5526. [[CrossRef](#)]
63. Dong, B.X.; Wang, L.Z.; Song, L.; Zhao, J.; Teng, Y.L. Thermochemical Reduction of Carbon Dioxide with Alkali Metal Hydrides, Producing Methane and Hydrogen Fuels at Moderate Temperatures. *Energy Fuels* **2016**, *30*, 6620–6625. [[CrossRef](#)]
64. Zhao, J.; Teng, Y.L.; Dong, B.X. Thermal Reduction of CO₂ with Activated Alkali Metal Aluminum Hydrides for Selective Methanation. *Energy Fuels* **2020**, *34*, 11210–11218. [[CrossRef](#)]
65. Nilwanna, K.; Sittiwong, J.; Boekfa, B.; Treesukol, P.; Boonya-udtayan, S.; Probst, M.; Maihom, T.; Limtrakul, J. Aluminum-based metal-organic framework support metal(II)-hydride as catalyst for the hydrogenation of carbon dioxide to formic acid: A computational study. *Mol. Catal.* **2023**, *541*, 113116. [[CrossRef](#)]
66. Zhang, X.; Liu, G.; Meiwe-Broer, K.H.; Gantefor, G.; Bowen, K. CO₂ Activation and Hydrogenation by PtHn[−] Cluster Anions. *Angew. Chemie—Int. Ed.* **2016**, *55*, 9644–9647. [[CrossRef](#)] [[PubMed](#)]
67. Habib, M.; Sarkar, R.; Biswas, S.; Pramanik, A.; Sarkar, P.; Pal, S. Unambiguous hydrogenation of CO₂ by coinage-metal hydride anions: An intuitive idea based on in silico experiments. *Phys. Chem. Chem. Phys.* **2019**, *21*, 7483–7490. [[CrossRef](#)] [[PubMed](#)]
68. Kadeer, K.; Li, X.; Zheng, J. Hydrogen generation by coupling methanol steam reforming with metal hydride hydrolysis. *Chem. Commun.* **2023**, *59*, 5443–5446. [[CrossRef](#)]
69. Kim, J.; Lim, H.; Tian, Y.; Pili, L.; Hunt, A.; Waluyo, I.; Senanayake, S.D.; Rodriguez, J.A. The Surface Chemistry of Methanol on Pd(111) and H-Pd(111) Surfaces: C-O Bond Cleavage and the Effects of Metal Hydride Formation. *J. Phys. Chem. Lett.* **2024**, *15*, 6209–6215. [[CrossRef](#)]

70. Rusman, N.A.A.; Dahari, M. A review on the current progress of metal hydrides material for solid-state hydrogen storage applications. *Int. J. Hydrogen Energy* **2016**, *41*, 12108–12126. [\[CrossRef\]](#)
71. Akamaru, S.; Hara, M.; Matsuyama, M. Alloying effects on the hydrogen-storage capability of Pd-TM-H (TM = Cu, Au, Pt, Ir) systems. *J. Alloys Compd.* **2014**, *614*, 238–243. [\[CrossRef\]](#)
72. Yao, J.; Wang, B.; Chen, H.; Han, Z.; Wu, Y.; Cai, Z.; Manggada, G.W.; Elsayed, M.A.; Zhou, S. Effect of copper cluster on reaction pathways of carbon dioxide hydrogenation on magnesium hydride surface. *Int. J. Hydrogen Energy*. **2024**, *78*, 1089–1098. [\[CrossRef\]](#)
73. Jin, P.; Luo, N.; Wang, F. Analogy in the Mechanism of Heterolytic H₂ Dissociation. *ACS Catal.* **2024**, *14*, 18639–18650. [\[CrossRef\]](#)
74. Kyriakou, G.; Boucher, M.B.; Jewell, A.D.; Lewis, E.A.; Lawton, T.J.; Baber, A.E.; Tierney, H.L.; Flytzani-Stephanopoulos, M.; Sykes, E.C.H. Isolated metal atom geometries as a strategy for selective heterogeneous hydrogenations. *Science* **2012**, *335*, 1209–1212. [\[CrossRef\]](#)
75. Yang, C.; Ma, S.; Liu, Y.; Wang, L.; Yuan, D.; Shao, W.; Zhang, L.; Yang, F.; Lin, T.; Ding, H.; et al. Homolytic H₂ dissociation for enhanced hydrogenation catalysis on oxides. *Nat. Commun.* **2024**, *15*, 540. [\[CrossRef\]](#) [\[PubMed\]](#)
76. Xiang, S.; Dong, L.; Wang, Z.; Han, X.; Daemen, L.L.; Li, J.; Cheng, Y.; Guo, Y.; Liu, X.; Hu, Y.; et al. A unique Co@CoO catalyst for hydrogenolysis of biomass-derived 5-hydroxymethylfurfural to 2,5-dimethylfuran. *Nat. Commun.* **2022**, *13*, 3657. [\[CrossRef\]](#)
77. Vilé, G.; Albani, D.; Nachtegaal, M.; Chen, Z.; Dontsova, D.; Antonietti, M.; López, N.; Pérez-Ramírez, J. A Stable Single-Site Palladium Catalyst for Hydrogenations. *Angew. Chemie—Int. Ed.* **2015**, *54*, 11265–11269. [\[CrossRef\]](#)
78. Eisenstein, O.; Crabtree, R.H. Outer sphere hydrogenation catalysis. *New J. Chem.* **2013**, *37*, 21–27. [\[CrossRef\]](#)
79. Aireddy, D.R.; Ding, K. Heterolytic Dissociation of H₂ in Heterogeneous Catalysis. *ACS Catal.* **2022**, *12*, 4707–4723. [\[CrossRef\]](#)
80. Wu, Z.; Cheng, Y.; Tao, F.; Daemen, L.; Foo, G.S.; Nguyen, L.; Zhang, X.; Beste, A.; Ramirez-Cuesta, A.J. Direct Neutron Spectroscopy Observation of Cerium Hydride Species on a Cerium Oxide Catalyst. *J. Am. Chem. Soc.* **2017**, *139*, 9721–9727. [\[CrossRef\]](#)
81. Strunk, J.; Kähler, K.; Xia, X.; Muhler, M. The surface chemistry of ZnO nanoparticles applied as heterogeneous catalysts in methanol synthesis. *Surf. Sci.* **2009**, *603*, 1776–1783. [\[CrossRef\]](#)
82. Ye, J.; Liu, C.; Mei, D.; Ge, Q. Active Oxygen Vacancy Site for Methanol Synthesis from CO₂ Hydrogenation on In₂O₃(110): A DFT Study. *Am. Chem. Soc.* **2013**, *3*, 1296–1306. [\[CrossRef\]](#)
83. Frei, M.S.; Capdevila-Cortada, M.; García-Muelas, R.; Mondello, C.; López, N.; Stewart, J.A.; Ferré, D.C.; Pérez-Ramírez, J. Mechanism and microkinetics of methanol synthesis via CO₂ hydrogenation on indium oxide. *J. Catal.* **2018**, *361*, 313–321. [\[CrossRef\]](#)
84. Qin, B.; Li, S. First principles investigation of dissociative adsorption of H₂ during CO₂ hydrogenation over cubic and hexagonal In₂O₃ catalysts. *Phys. Chem. Chem. Phys.* **2020**, *22*, 3390–3399. [\[CrossRef\]](#)
85. Jia, J.; Qian, C.; Dong, Y.; Li, Y.F.; Wang, H.; Ghoussoub, M.; Butler, K.T.; Walsh, A.; Ozin, G.A. Heterogeneous catalytic hydrogenation of CO₂ by metal oxides: Defect engineering-perfecting imperfection. *Chem. Soc. Rev.* **2017**, *46*, 4631–4644. [\[CrossRef\]](#)
86. Posada-Borbón, A.; Grönbeck, H. Hydrogen adsorption on In₂O₃(111) and In₂O₃(110). *Phys. Chem. Chem. Phys.* **2020**, *22*, 16193–16202. [\[CrossRef\]](#)
87. Ye, J.; Ge, Q.; Liu, C. DFT study of CO₂ adsorption and hydrogenation on the In₂O₃ surfaces. *J. Phys. Chem.* **2012**, *116*, 7817–7825. [\[CrossRef\]](#)
88. Juárez, R.; Parker, S.F.; Concepción, P.; Corma, A.; García, H. Heterolytic and heterotopic dissociation of hydrogen on ceria-supported gold nanoparticles. Combined inelastic neutron scattering and FT-IR spectroscopic study on the nature and reactivity of surface hydrogen species. *Chem. Sci.* **2010**, *1*, 731–738. [\[CrossRef\]](#)
89. Mao, G.C.; Kan, X.T.; Xiao, M.X.; Liu, W.L.; Dong, B.X.; Teng, Y.L. Alkaline Earth Metal-Induced Hydrogenation of the CaO-Captured CO₂ to Methane at Room Temperature. *Ind. Eng. Chem. Res.* **2022**, *61*, 10124–10132. [\[CrossRef\]](#)
90. Tang, Y.; Kobayashi, Y.; Tassel, C.; Yamamoto, T.; Kageyama, H. Hydride-Enhanced CO₂ Methanation: Water-Stable BaTiO_{2.4}H_{0.6} as a New Support. *Adv. Energy Mater.* **2018**, *8*, 1800800. [\[CrossRef\]](#)
91. He, Y.; Li, Y.; Lei, M.; Polo-Garzon, F.; Perez-Aguilar, J.; Bare, S.R.; Formo, E.; Kim, H.; Daemen, L.; Cheng, Y.; et al. Significant Roles of Surface Hydrides in Enhancing the Performance of Cu/BaTiO_{2.8}H_{0.2} Catalyst for CO₂ Hydrogenation to Methanol. *Angew. Chemie—Int. Ed.* **2024**, *63*, e202313389. [\[CrossRef\]](#)
92. Grice, K.A. Carbon dioxide reduction with homogenous early transition metal complexes: Opportunities and challenges for developing CO₂ catalysis. *Coord. Chem. Rev.* **2017**, *336*, 78–95. [\[CrossRef\]](#)
93. Das, C.; Grover, J.; Tannu, Das, A.; Maiti, D.; Dutta, A.; Lahiri, G.K. Recent developments in first-row transition metal complex-catalyzed CO₂ hydrogenation. *Dalt. Trans.* **2022**, *51*, 8160–8168. [\[CrossRef\]](#) [\[PubMed\]](#)
94. Fong, H.; Peters, J.C. Hydricity of an Fe-H species and catalytic CO₂ hydrogenation. *Inorg. Chem.* **2015**, *54*, 5124–5135. [\[CrossRef\]](#) [\[PubMed\]](#)

95. Ramakrishnan, S.; Waldie, K.M.; Warnke, I.; De Crisci, A.G.; Batista, V.S.; Waymouth, R.M.; Chidsey, C.E.D. Experimental and Theoretical Study of CO₂ Insertion into Ruthenium Hydride Complexes. *Inorg. Chem.* **2016**, *55*, 1623–1632. [[CrossRef](#)]
96. Hadlington, T.J.; Kefalidis, C.E.; Maron, L.; Jones, C. Efficient Reduction of Carbon Dioxide to Methanol Equivalents Catalyzed by Two-Coordinate Amido-Germanium(II) and -Tin(II) Hydride Complexes. *ACS Catal.* **2017**, *7*, 1853–1859. [[CrossRef](#)]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.